

CTS™ Rotea™ Counterflow Centrifugation System SmartStart Training

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Presentation Outline

1 Rotea Instrument and Single Use Kit

2 Introduction to Counterflow Centrifugation

3 Basic Protocol Steps

4 Troubleshooting

5 Technical Support

Presentation Outline

1 Rotea Instrument and Single Use Kit

2 Introduction to Counterflow Centrifugation

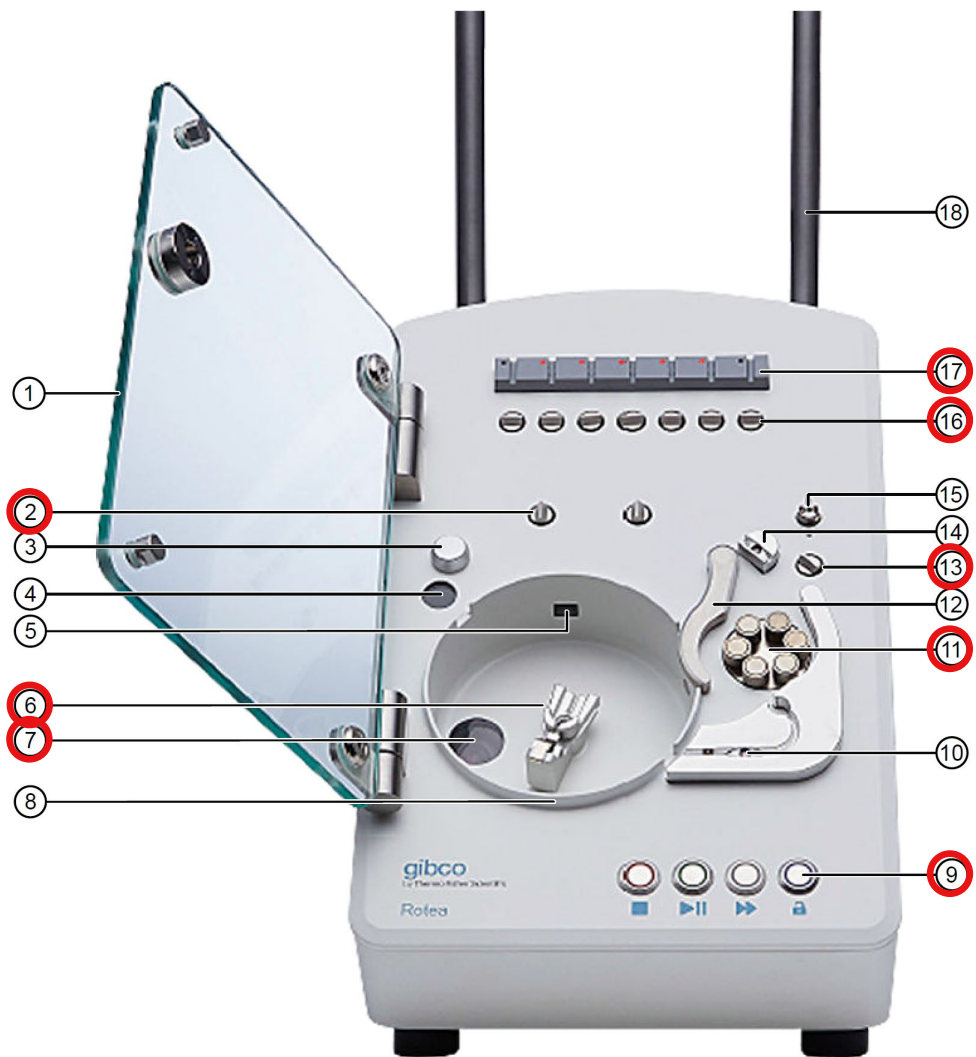
3 Basic Protocol Steps

4 Troubleshooting

5 Technical Support



Rotea Instrument and Single Use Kit



- 1) Glass door
- 2) Pinch valves J & K
- 3) Kit location button
- 4) 2D barcode reader
- 5) CFC chamber detector
- 6) CFC chamber carrier
- 7) Camera
- 8) Moisture Sensor
- 9) Push button controls
- 10) OD sensor and P1 pressure sensor
- 11) Peristaltic pump
- 12) Pump clamp arm
- 13) Pinch valve H
- 14) Kit location button and P2 pressure sensor
- 15) Door latch
- 16) Pinch valves A-G
- 17) Bubble sensors B-F
- 18) Hanger posts

Rotea Instrument and Single Use Kit

Stop: stops current protocol immediately

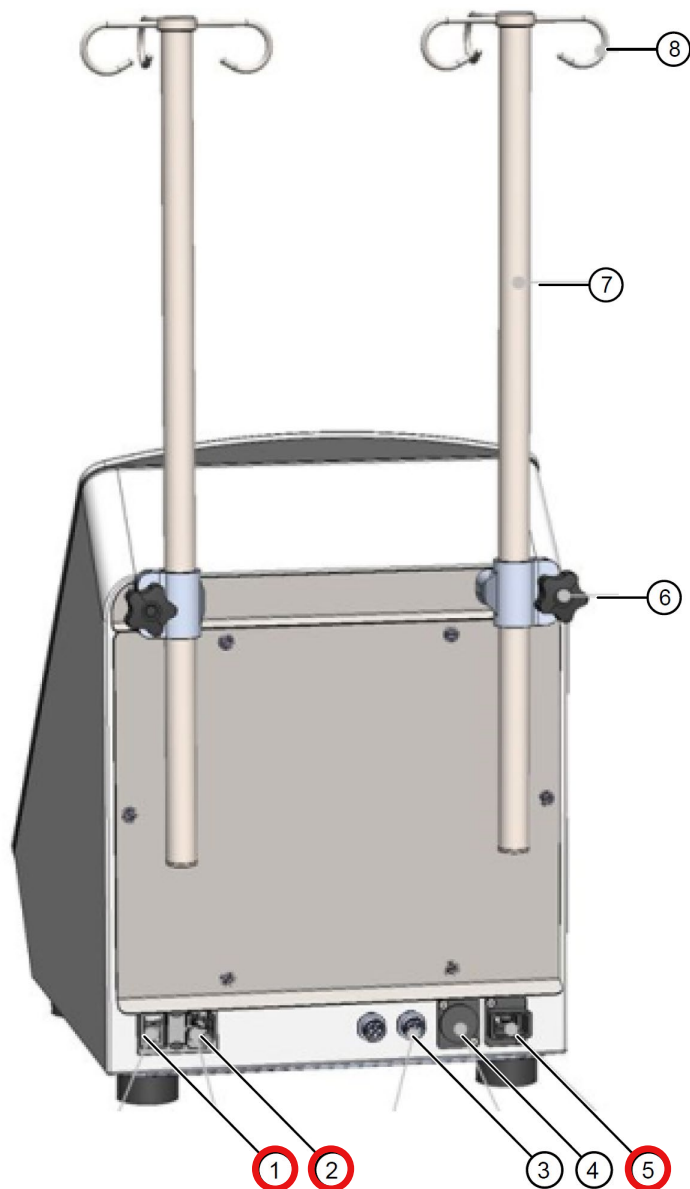


Door unlock: retracts valves, then opens door

Play/pause: initially starts protocol, pauses a protocol in progress, resumes protocol

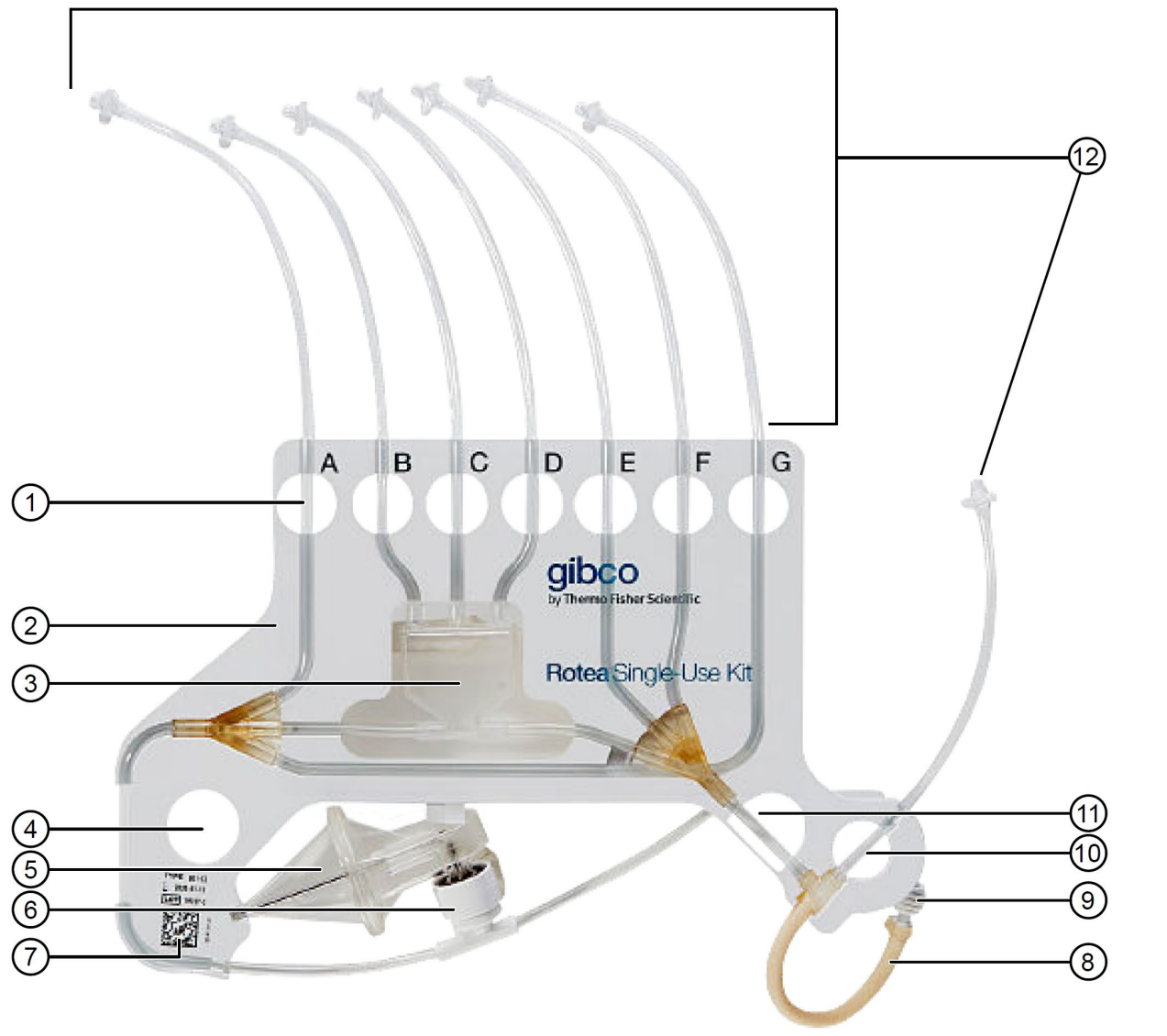
Advance: skips to next step in protocol, if enabled (solid white light)

Rotea Instrument and Single Use Kit



- 1) Power switch
- 2) Power connection
- 3) CAN bus
- 4) Ethernet (RJ-45)
- 5) USB-C
- 6) Hanger post adjustment knobs
- 7) Hanger post
- 8) Bag hooks

Rotea Instrument and Single Use Kit



- 1) Valve hole
- 2) Carrier frame
- 3) Bubble trap
- 4) Location hole
- 5) CFC chamber
- 6) Rotary coupling
- 7) 2D barcode
- 8) Pump tubing
- 9) Tube retainer
- 10) Valve H hole
- 11) Location hole

New Full Flow Single-Use Kit (Type 004)



Components	Full Flow Single-Use Kits	Standard Single-Use Kits	Hi-Flow Single-Use Kits
Pump tubing	PharmaPure Pump tubing	Bioprene pump tubing	Bioprene pump tubing
Barbed Tee	Tritan MX711 copolyester	Polypropylene	Polypropylene
PVC tube	Tygon PVC tube	Sunlite PVC tube	Sunlite PVC tube
O-ring	Viton	EPDM	EPDM
Tubing Diameter	OD: 4 mm (0.16") ID: 2.4 mm (0.94")	OD: 4 mm (0.16") ID: 3 mm (0.116")	OD: 4 mm (0.16") ID: 3 mm (0.116")
Flow rate	6 - 150 mL/min	5 - 110 mL/min	30 - 160 mL/min

Presentation Outline

1

Rotea Instrument and Single Use Kit

2

Introduction to Counterflow Centrifugation

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Basic Protocol Steps

4

Troubleshooting

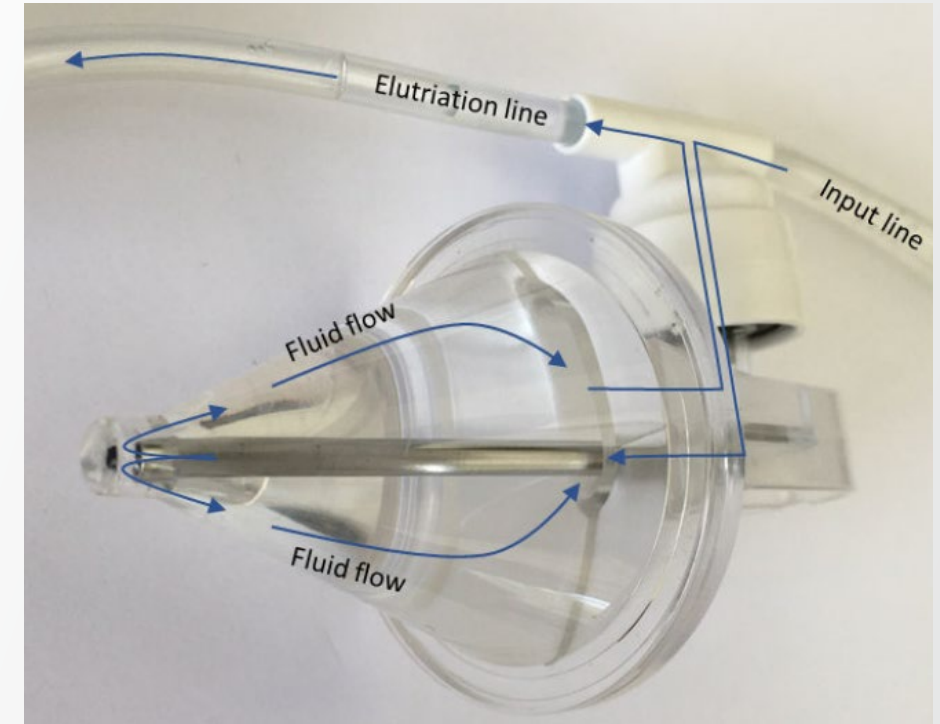
5

Technical Support

Introduction to Counterflow Centrifugation

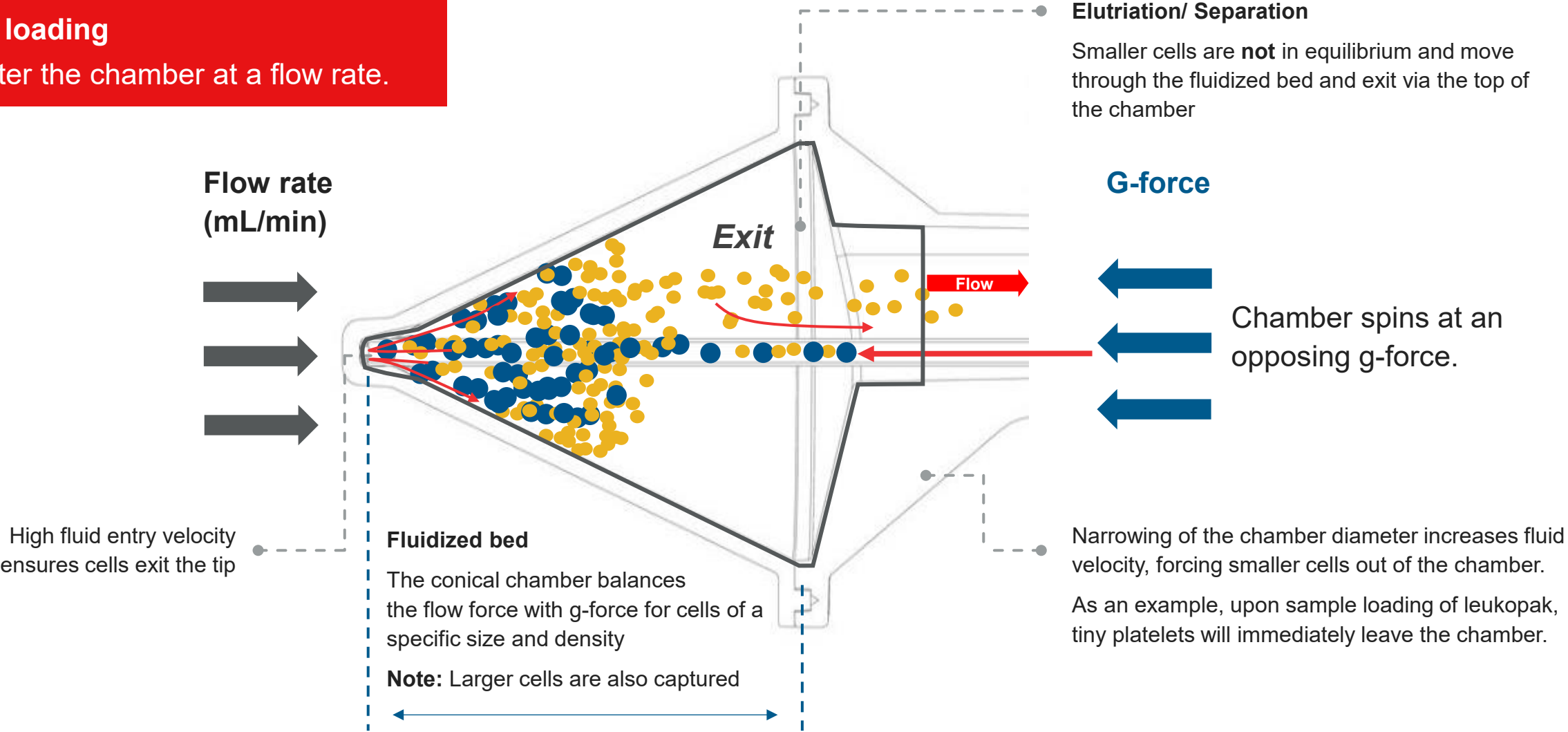
The position of cells/particles can be manipulated in the Counterflow Centrifugation (CFC) Chamber by adjusting (2) opposing forces:

- Force_{Centripetal} – Force due to rotation of CFC Chamber
- Force_{Stokes Drag} – Frictional force experienced by particles in a flowing viscous fluid



Introduction to Counterflow Centrifugation

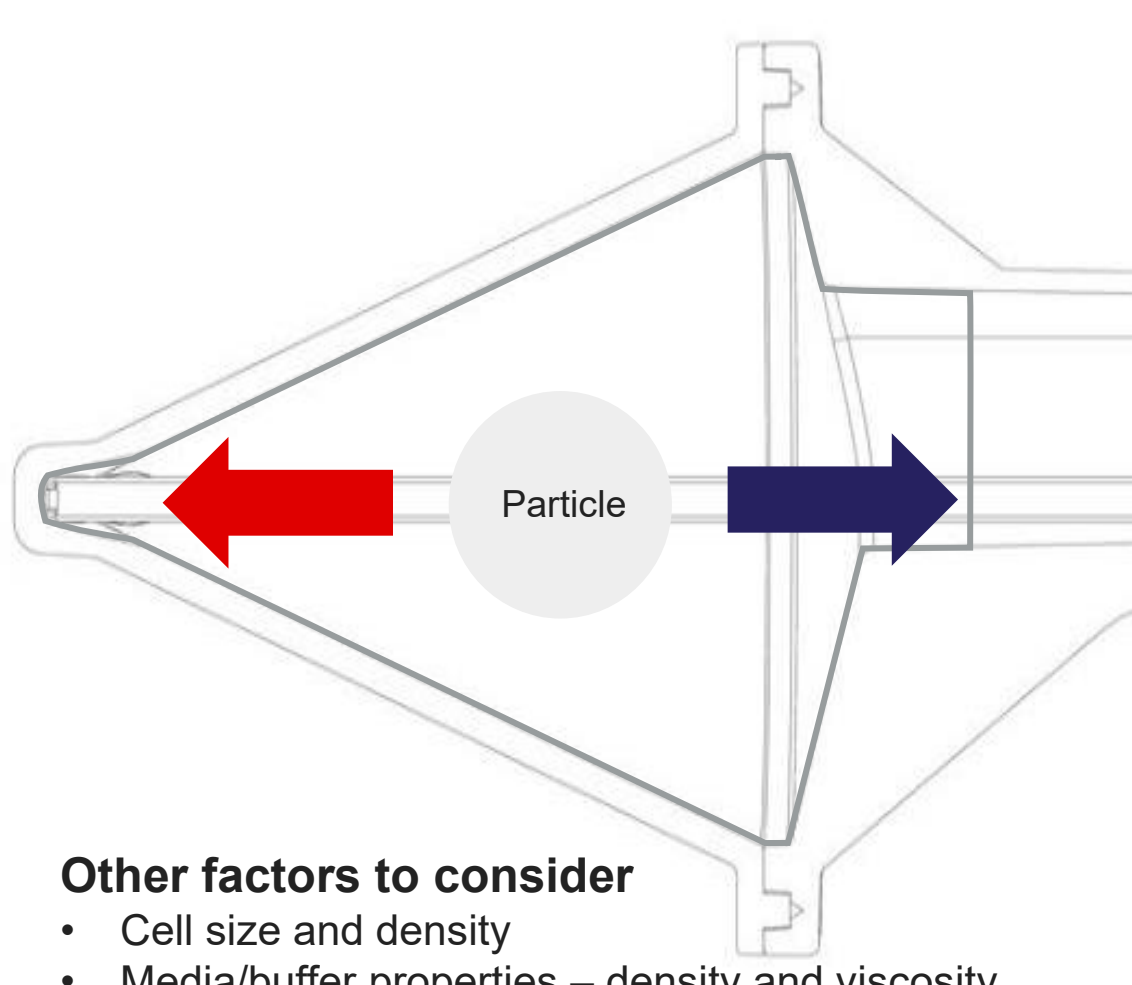
Sample loading
Cells enter the chamber at a flow rate.



Introduction to Counterflow Centrifugation

Remain in chamber

Raise centrifuge speed
and/or
Lower flow rate



Elutriate from chamber

Raise flow rate
and/or
Lower centrifuge speed

Other factors to consider

- Cell size and density
- Media/buffer properties – density and viscosity
- Cell concentration of input
- Homogeneous vs heterogeneous cell population

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Rotea GUI System Software Version – 1.7.30

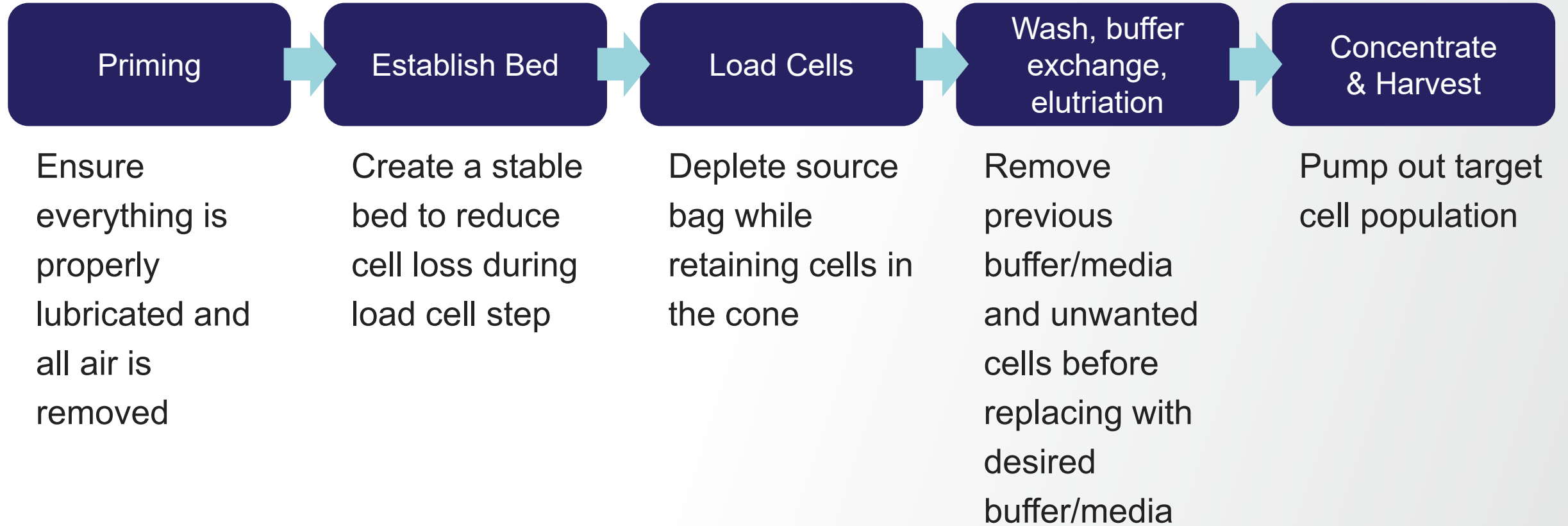
Rotea Protocol Builder Software – 2.0.16

Rotea SAE Software – 2.2

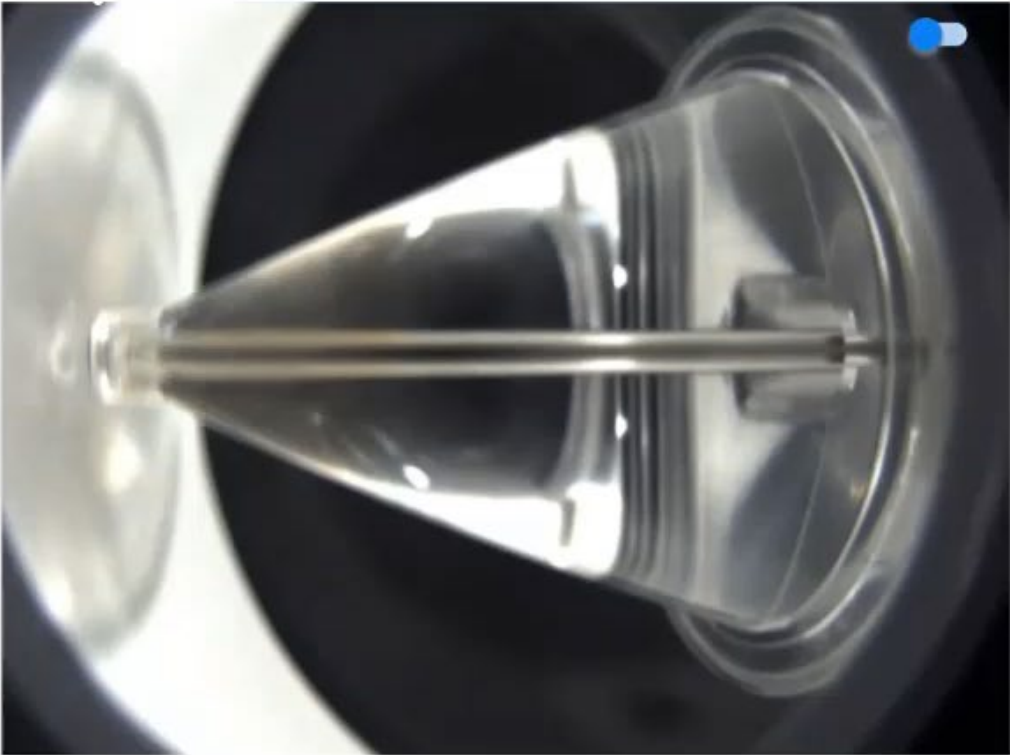
Rotea Firmware (main) - 0.75.26

Rotea Firmware (power) - 0.20.4

Basic Protocol Steps



Establish Bed



- 7. Prime Recirc
- 8. Ramp Speed
- ▶ 9. Establish Bed
 - 10. Load cells
 - 11. Ramp speeds to wash

More Steps

Instrument Status: Running

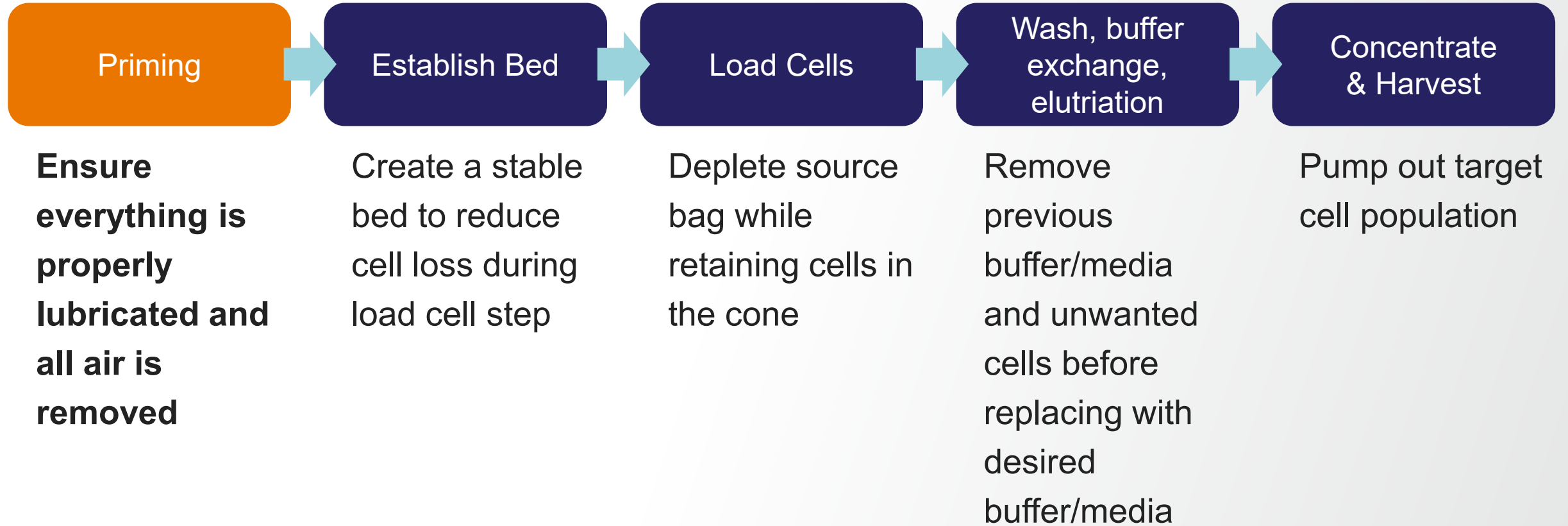
Total Time Remaining

Step Time Remaining 82 

Pump (mL/min) 25  

Centrifuge (g) 2500  

Basic Protocol Steps



Priming

The first steps in any protocol are used to prime the system.

Priming has two main functions:

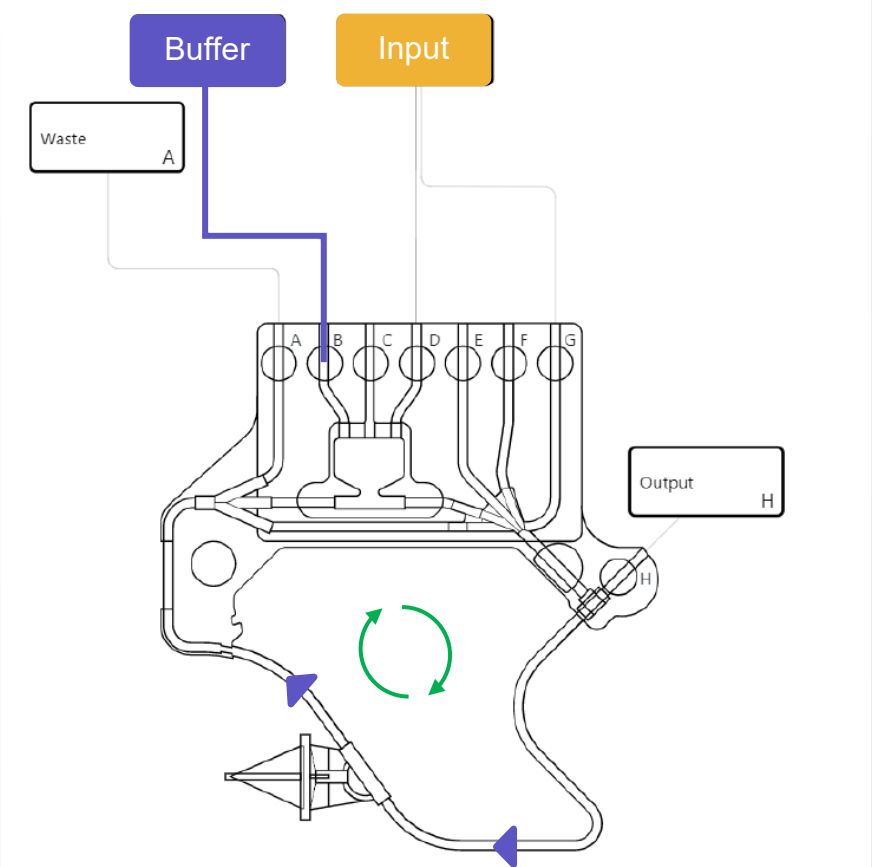
- Lubricate the Rotary Coupling
- Displace air in the kit

It is essential to prime any line through which fluid will enter the kit

Description	Prime volume	Kit
CFC Chamber (Cone + Base)	13.2 mL	
Bubble Trap	5 mL	
1/8" Tube (100 mm)	0.7 mL	
CFC Chamber to H	1.6 mL	
CFC Chamber to Bubble trap	2.4 mL	
CFC Chamber to E	2.6 mL	
CFC Chamber to F	2.6 mL	
CFC Chamber to A	1.7 mL	
Bubble Trap to B, C & D	0.3 mL	

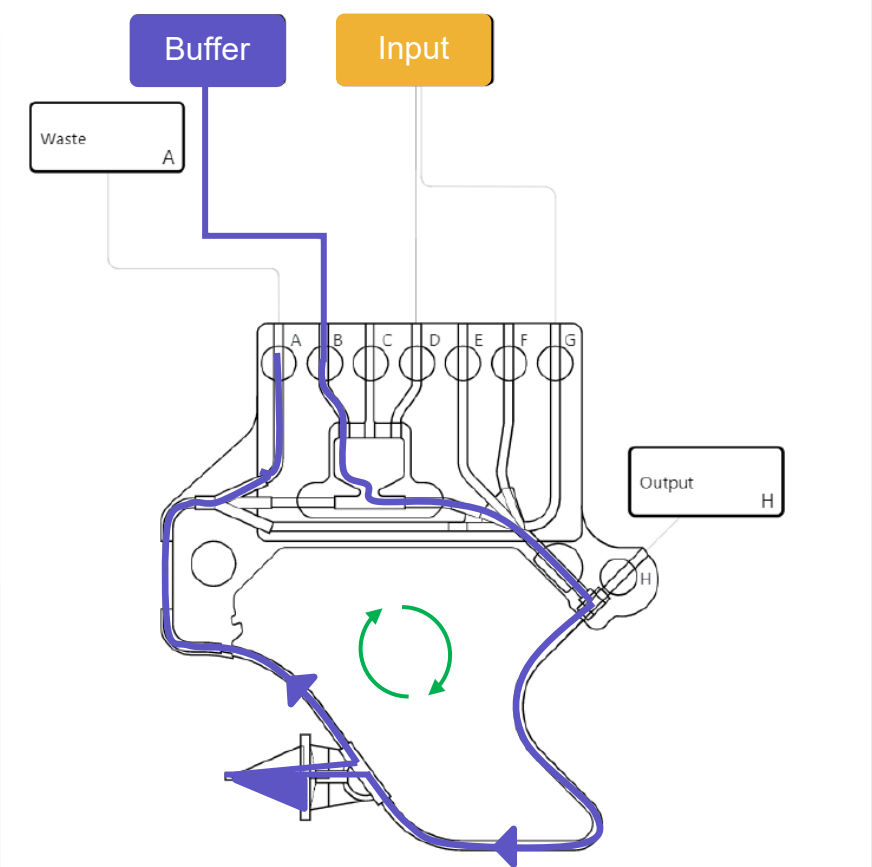
Priming

Step	Description
1	Pre-prime
2	Lubricate Rotary Coupling
3	Prime Chamber and line A
4	Add priming volume
5	Prime Bubble Trap and line B
6	Prime line D
7	Pressure prime
8	Prime Pause
9	Ramp speed to load



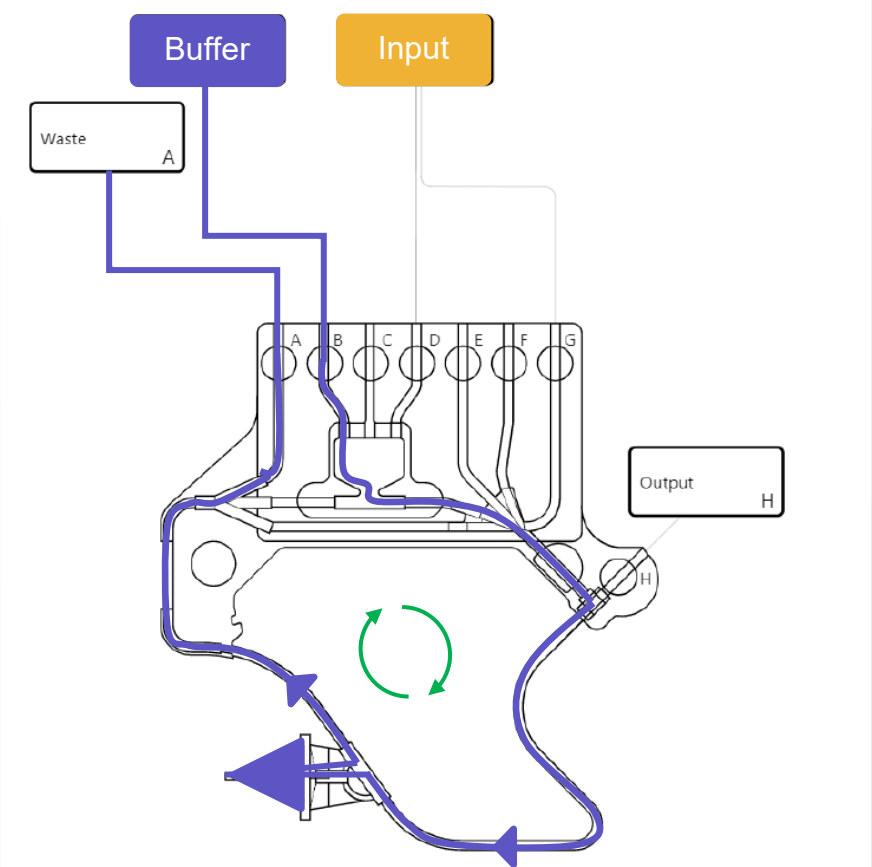
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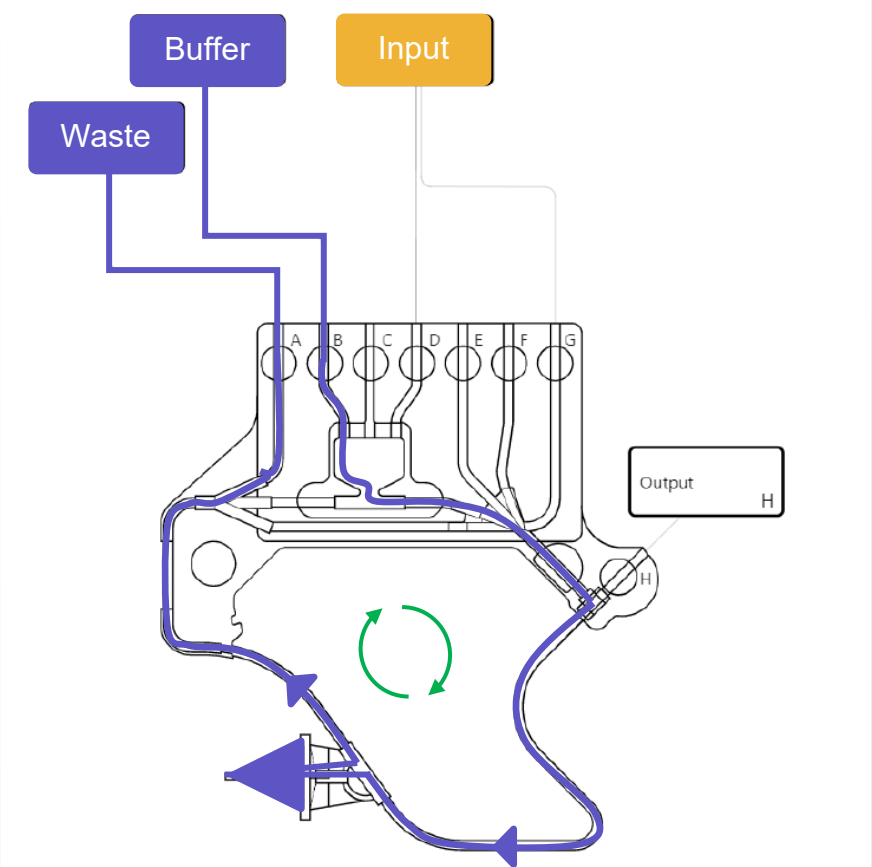
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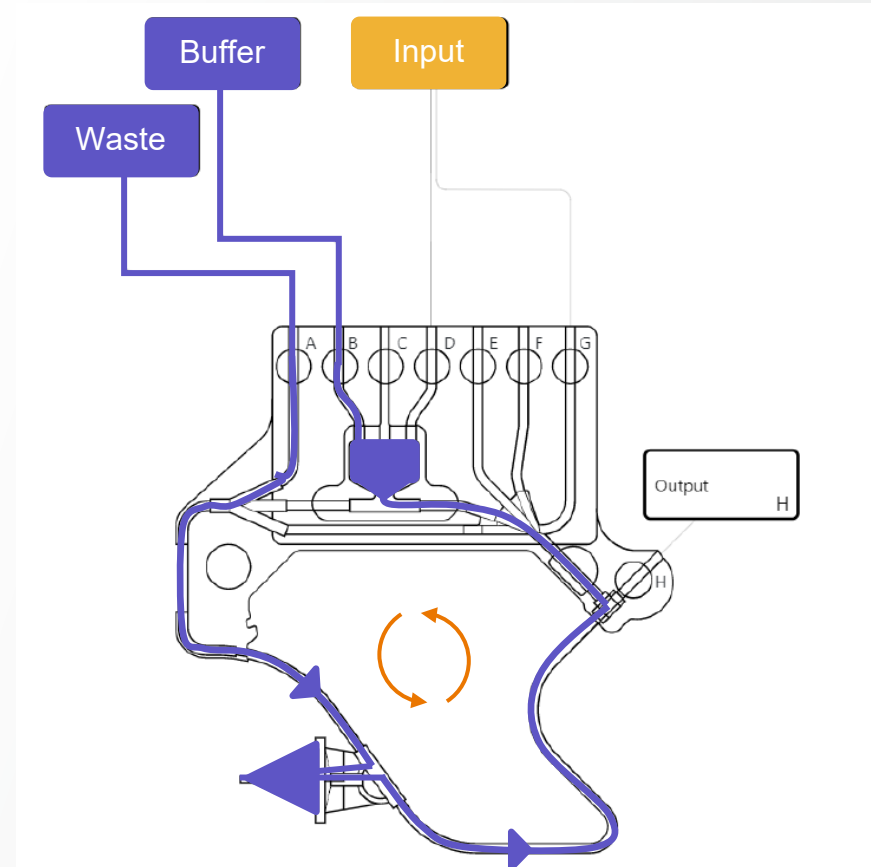
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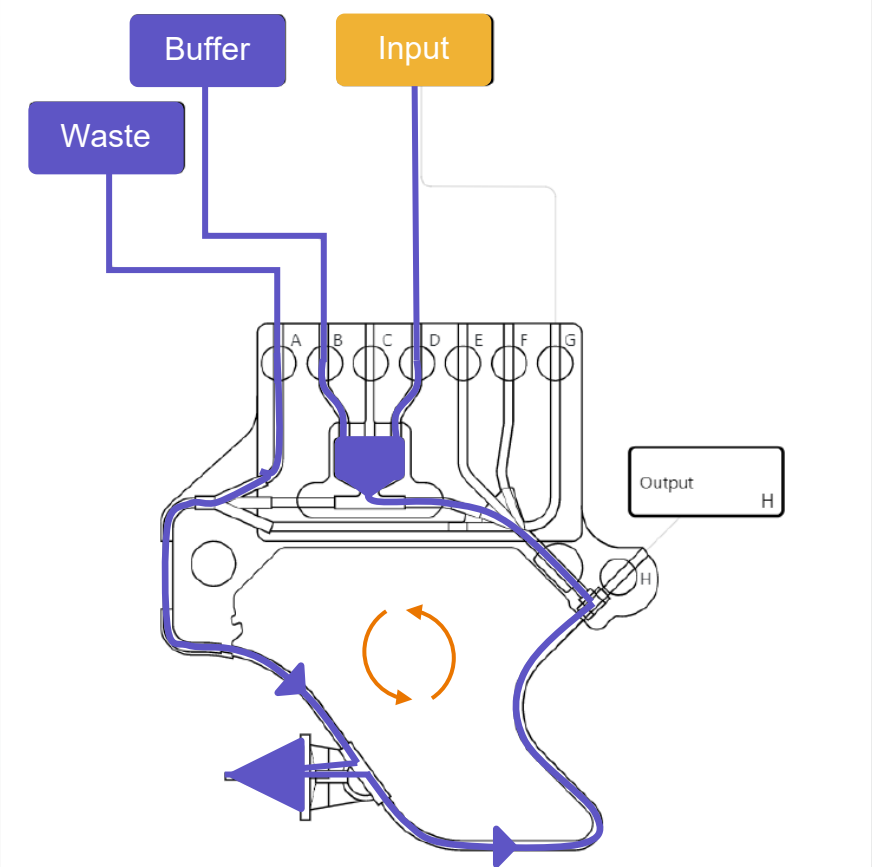
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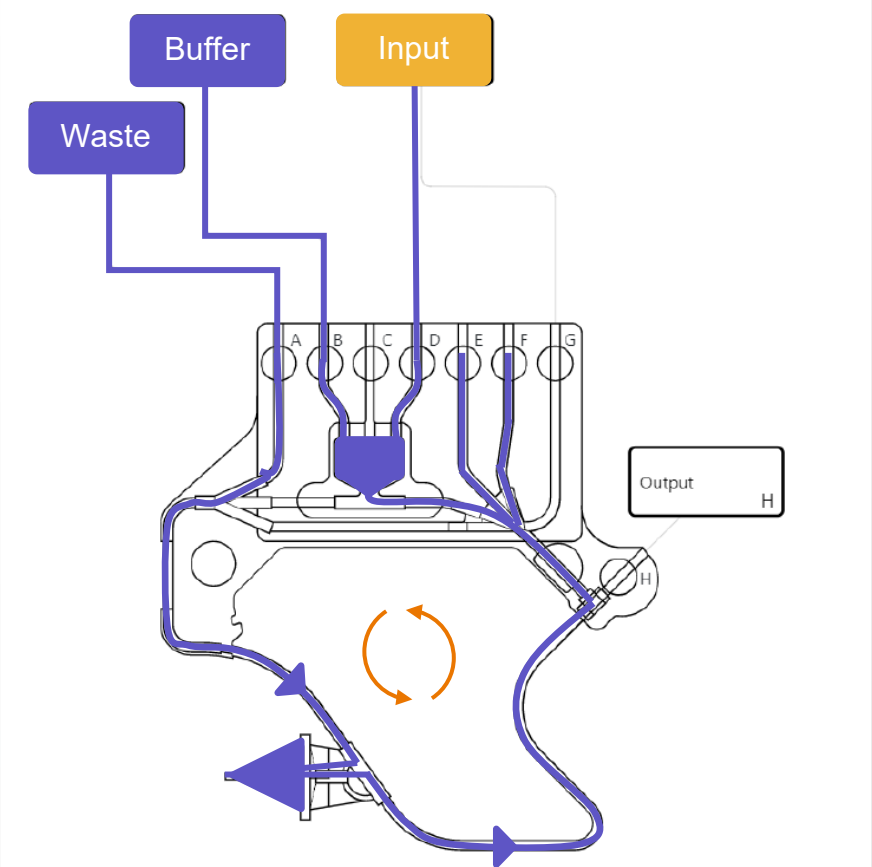
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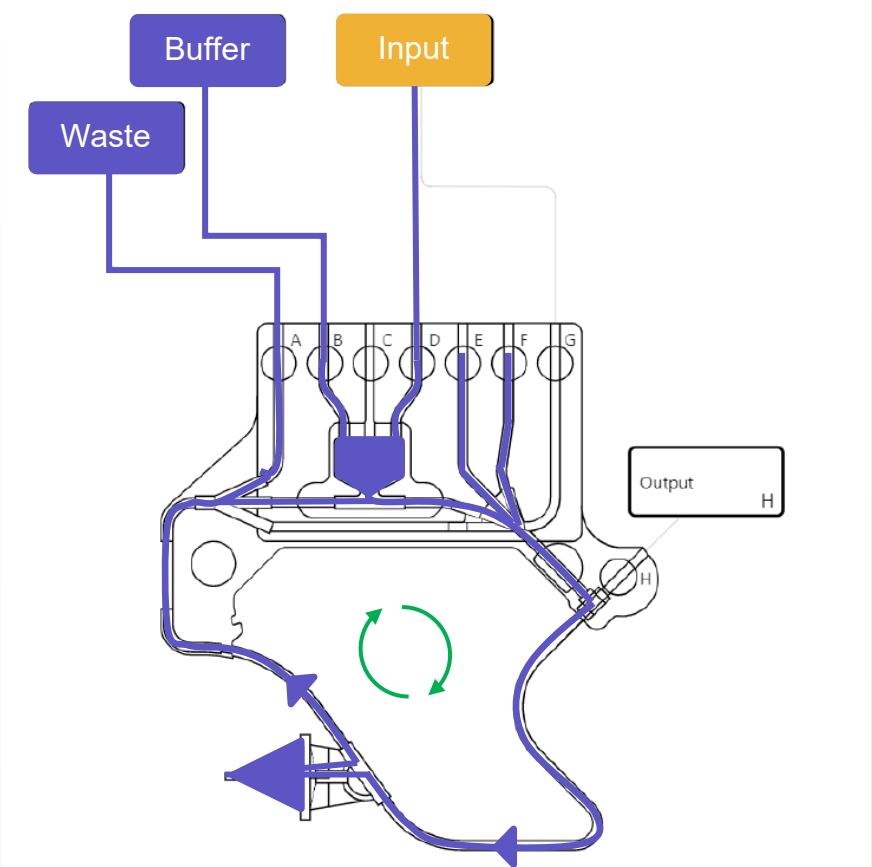
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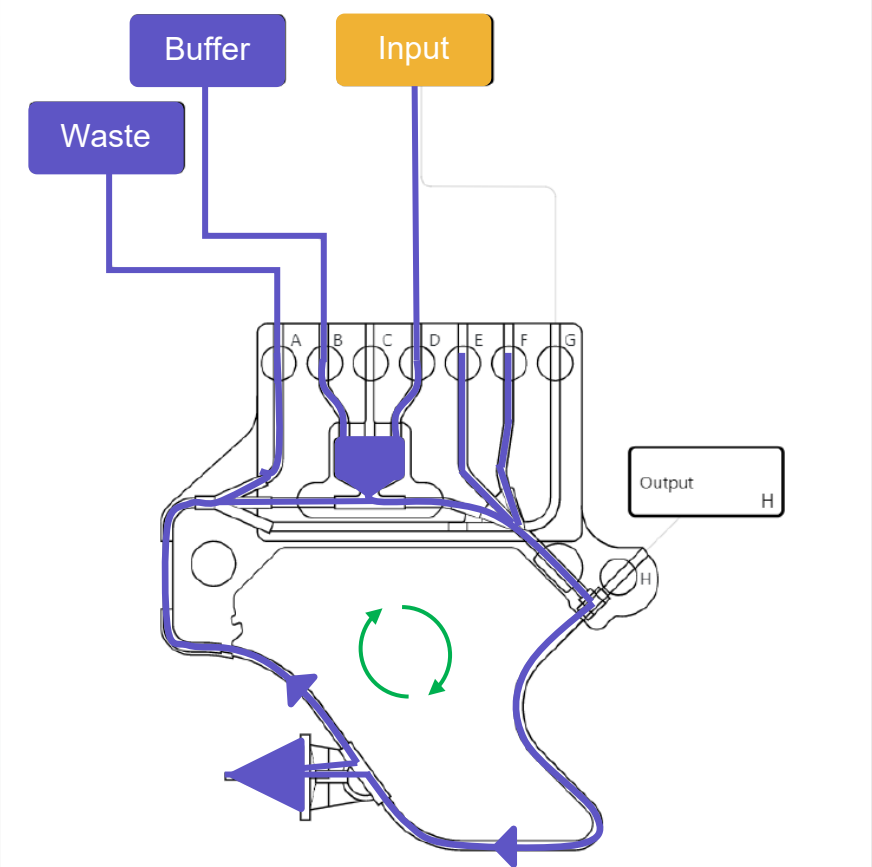
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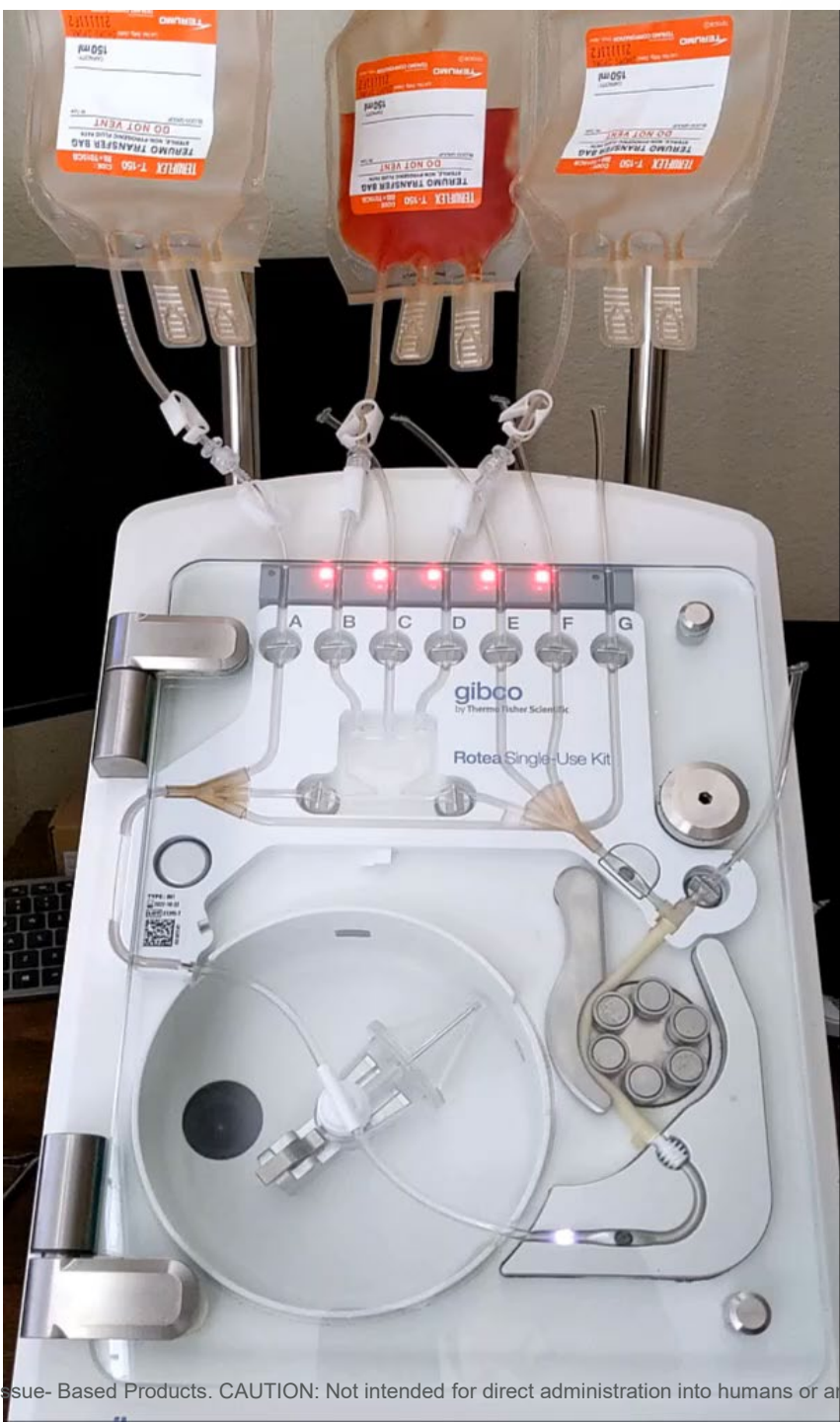


Priming

Step	Description
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6	Prime line D
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8	Prime Pause
9	Ramp speed to load



Priming



Pre-prime

Adjust for your specific protocol

- Make sure all the lines are primed, except harvest lines (that would add additional unwanted volume)
- If your lines are longer, you will have to increase the volume

Note: The priming sequence will always be protocol dependent. It is the user's responsibility to determine the priming steps that are appropriate to their specific application, the basic steps and settings shown (to the right) are recommended.

CTS™ Rotea™ Counterflow Centrifugation System Process Design USER GUIDE

A typical priming sequence consists of 9 steps, as shown in this example.

gibco
by Thermo Fisher Scientific

DIZ-TST-0026-B3-Standard Priming - LIST

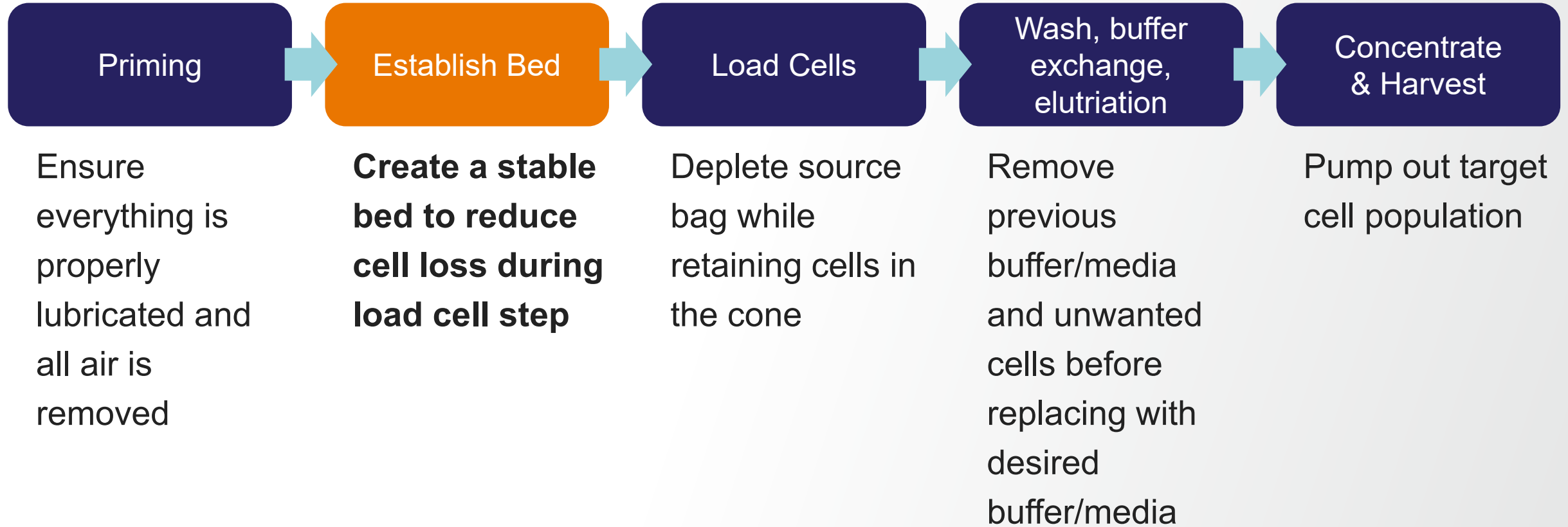
Kit List Data Entry 1 2 3 4 5 6 7 8 9

Summary: Standard Priming steps

Step	Description	Flow Path	Speed	Flow Rate	Step Type	Triggers
1	Pre-Prime	B to A	0g	100ml/min	Normal	Input Bubble Sensor
2	Lubricate Rotary Coupling	B to A	0g	100ml/min	Normal	Volume: 15 ml
3	Prime Chamber and Line A	B to A	10g	100ml/min	Normal	Volume: 15 ml
4	Add Priming Volume	B to A	10g	100ml/min	Normal	Volume: 20 ml
5	Prime Bubble Trap and Line B	A to B	10g	100ml/min	Normal	Volume: 15 ml
6	Prime Line D	A to D	10g	50ml/min	Normal	Volume: 5 ml
7	Pressure Prime	A to EF	10g	0ml/min	Pressure Prime	
8	Prime Pause Loop	J to K	10g	25ml/min	Pause	Volume: 3 ml
9	Ramp Speeds for Loading	J to K	2500g	40ml/min	Pause	Time: 10 seconds

Copy To Move To Insert Repeat Add Label Delete

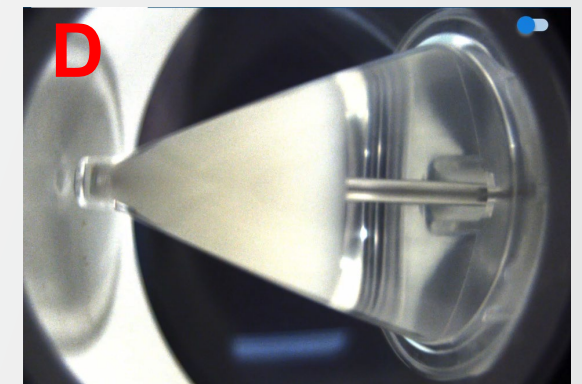
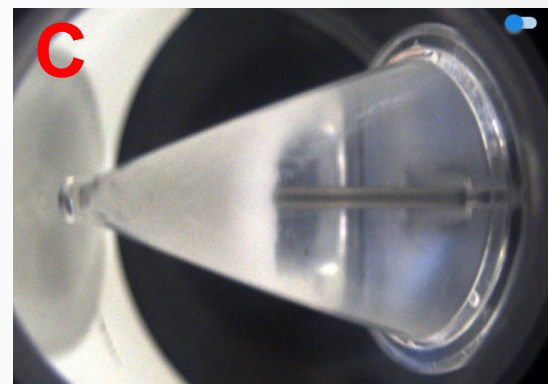
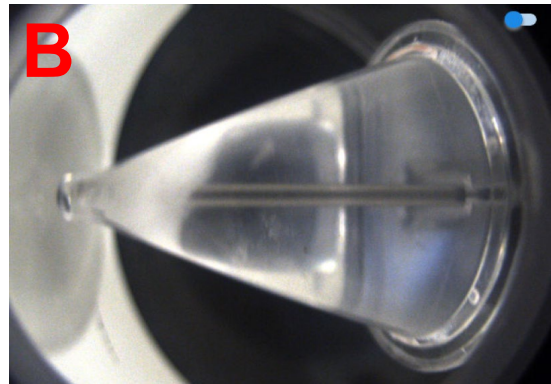
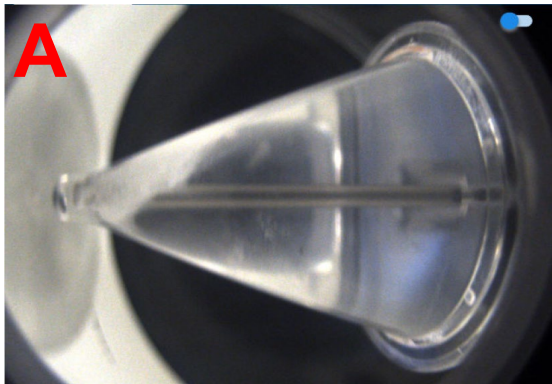
Basic Protocol Steps



Establish Bed

Stages of Bed formation

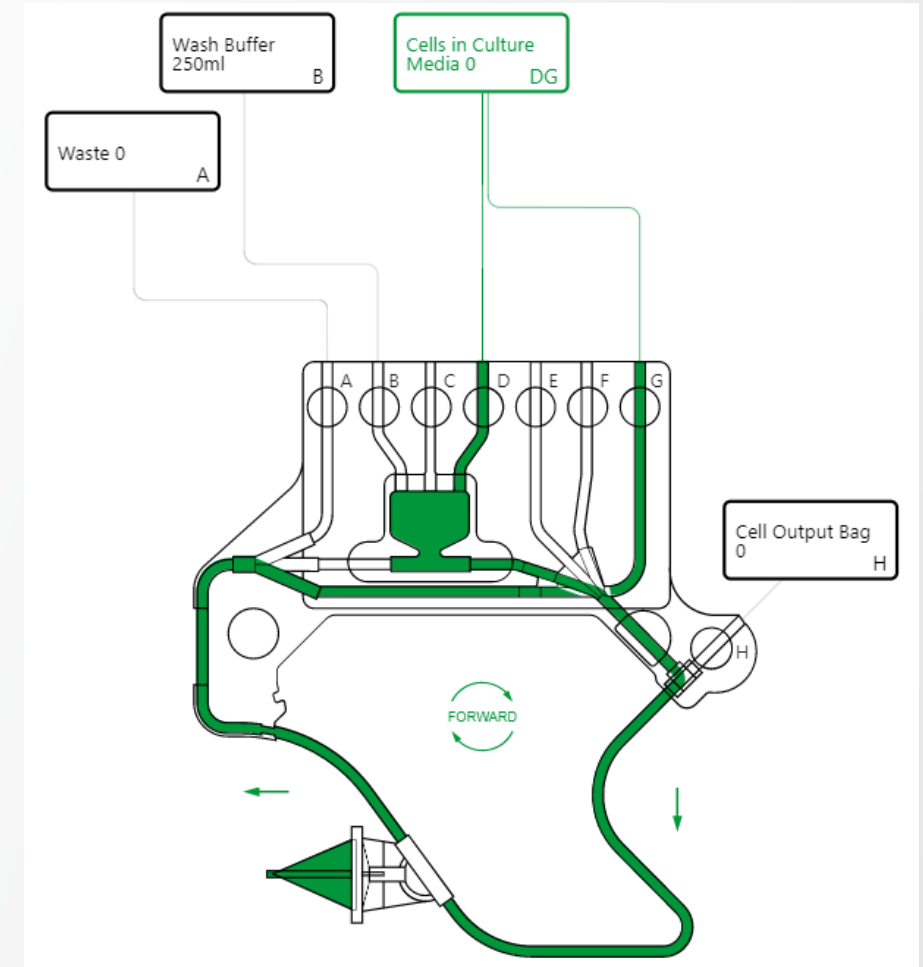
- In the early stages of establishing a bed (**A**), cells can be seen entering the chamber through the tip, and flowing up the top edge of the chamber, a consequence of Coriolis flow.
- As more cells enter the chamber, the bed begins to stabilize (**B-C**).
- A stable bed (**D**) will have a vertical bed interface with minimal turbulence.
 - Cells are not flowing up the top edge of the cone
 - The stainless-steel dip tube is visible at the tip of the CFC chamber
- When a stable cell bed forms depends on:
 - Cell type
 - Size of the bed
 - Media density



Establish Bed

Recirculation

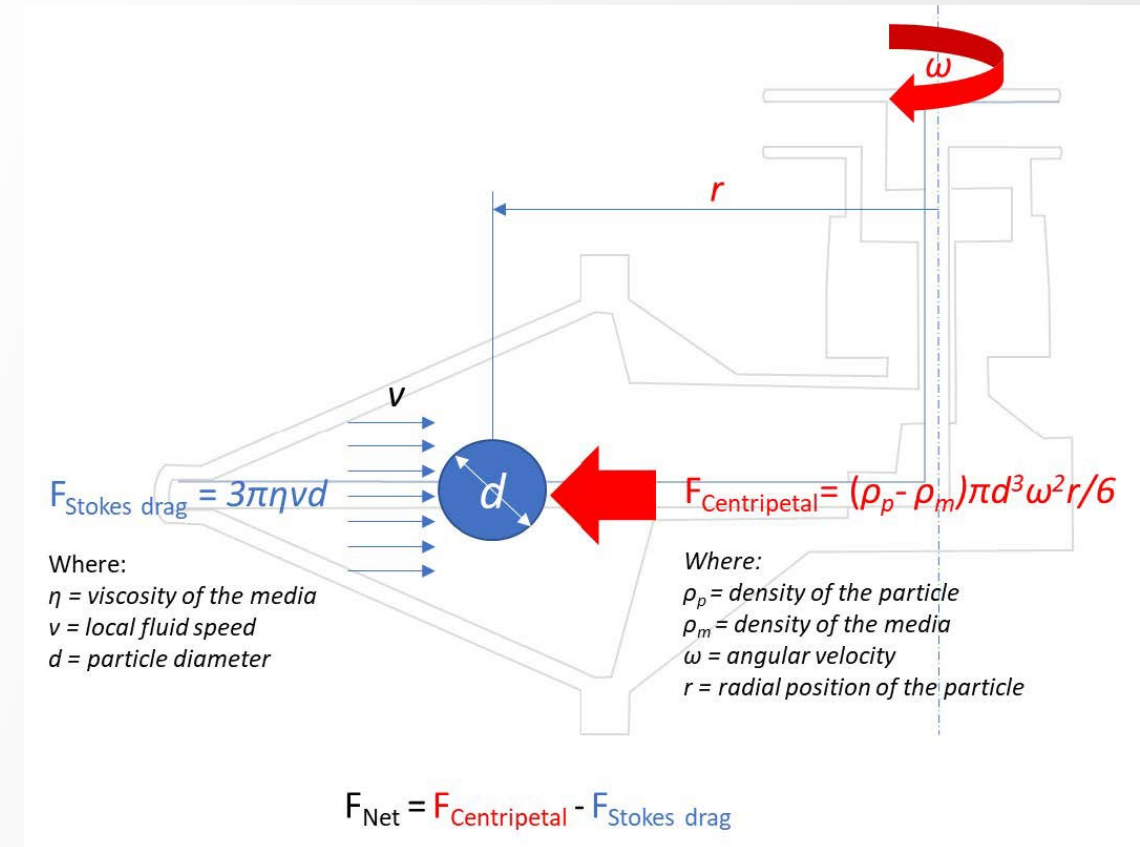
- Establishing a bed in recirculation ensures no cells are lost to the waste bag during the formation of the bed.
- The number of cells required for a fluidized bed to form is strongly influenced by the cell diameter and the density differential (between the media and cell density) but in general:
 - PBMCs: >30 million
 - MSCs: >20 million
- **It is best to work with larger numbers of cells**



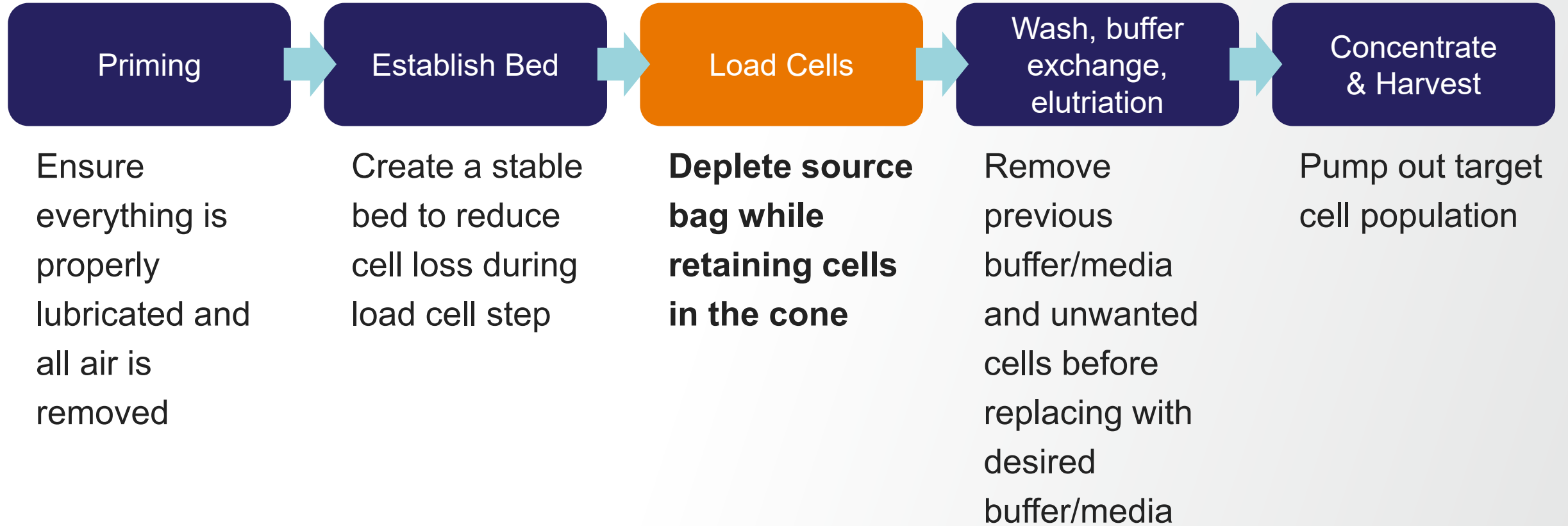
Establish Bed

A stable fluidized bed is the central concept that Rotea™ instrument is built around.

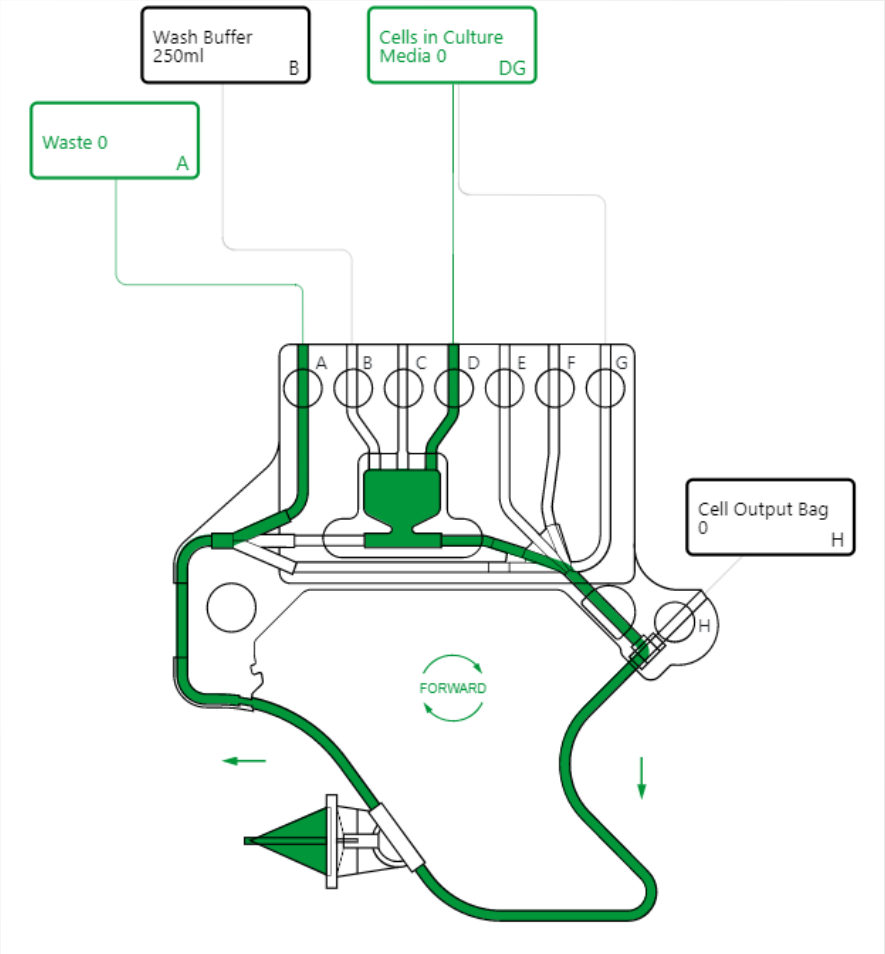
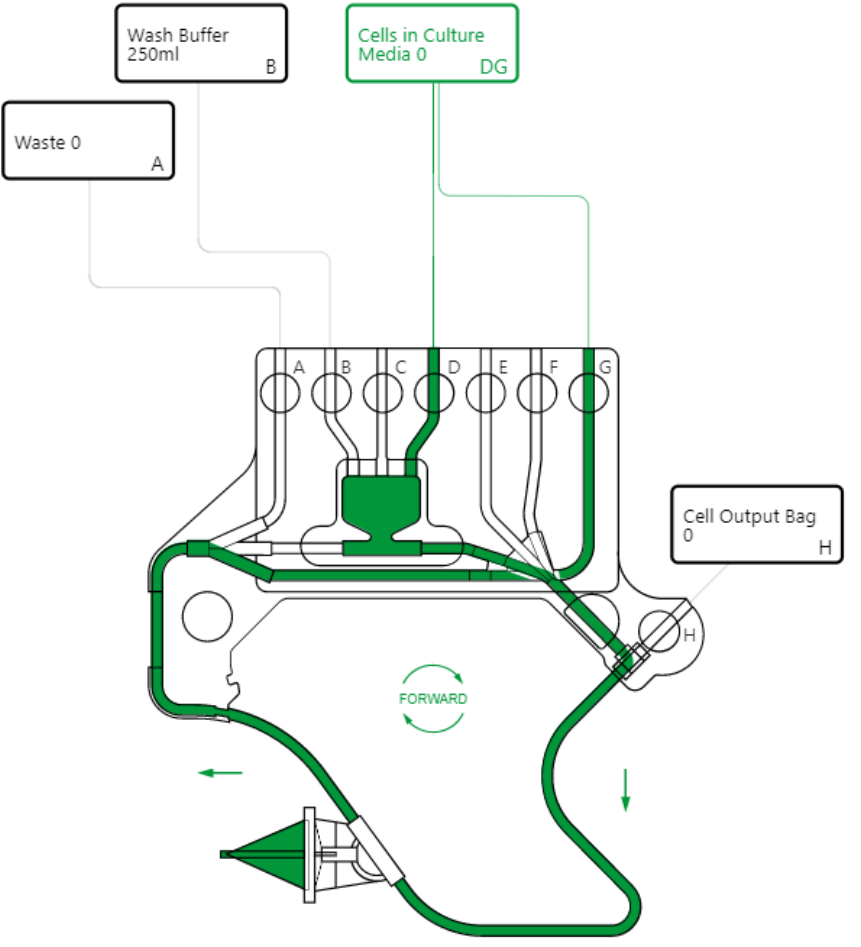
- Seven parameters affect formation of a fluidized bed to varying degrees and need to be considered when developing a protocol.
 - Cell diameter (1) and cell density (2) are fixed for the cell type.
 - G-force (3) and flow rate (4) are readily adjustable settings in the CTS™ Rotea™ Counterflow Centrifugation System.
 - Density (5) and viscosity (6) of media can sometimes be optimized to suit the protocol.
 - **Cell concentration (7) does not affect what settings are used but has a significant influence on the ability to form a bed.**



Basic Protocol steps



Load cells



Load cells

Process of capturing target cells in the fluidized bed as the input material passes through the Chamber

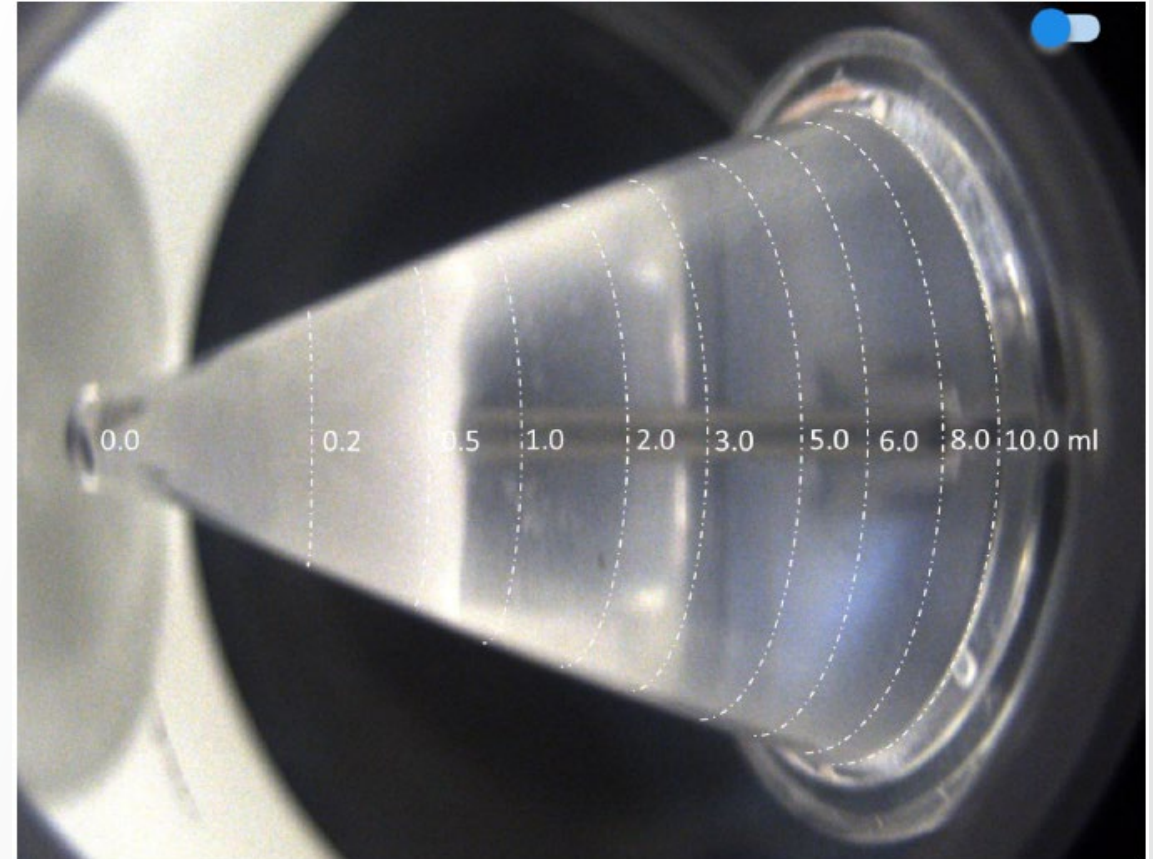
The number of cells that can be processed at a time is dependent on the following:

- Size and concentration of the cells
- Density of the media
- G-force and flow rate settings that will deliver acceptable cell recovery and viability

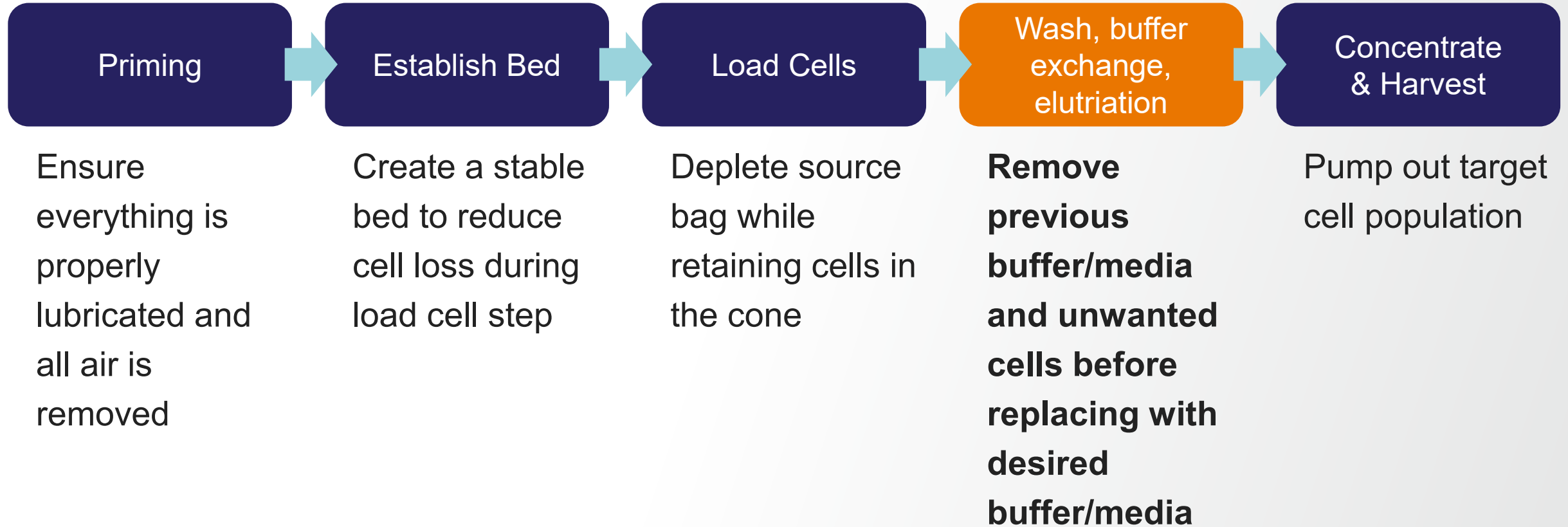
Generally, 2-3 billion T-cells is a good number of cells to work with at one time.

Cells $<4\text{ }\mu\text{m}$, e.g. platelets, cannot be retained in the cone.

Be sure to allow enough room in the cone so that your cells won't leave during the rest of the protocol.



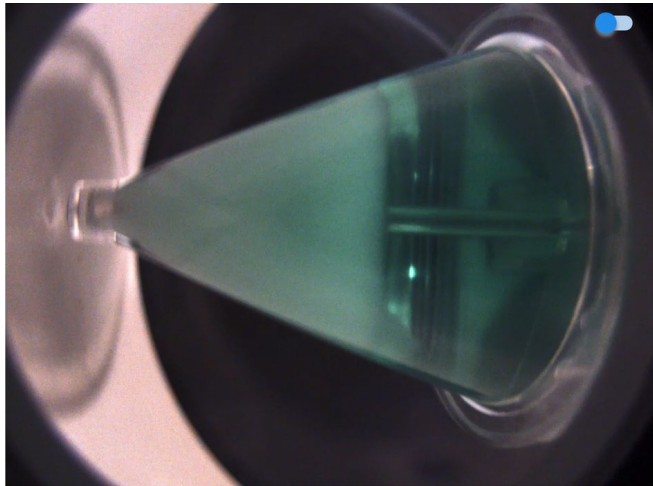
Basic Protocol steps



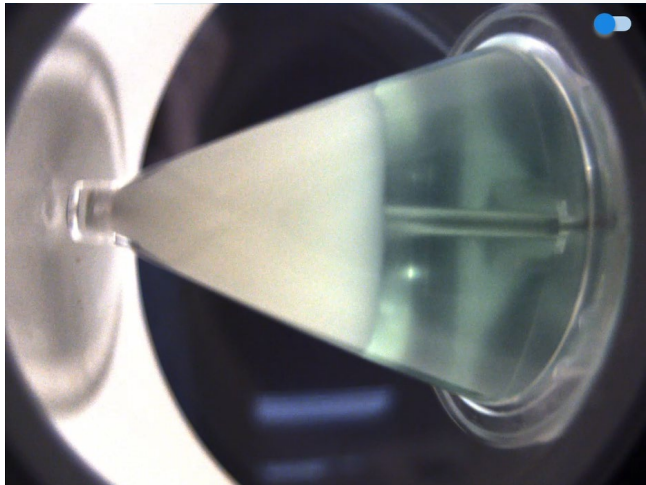
Basic Protocol steps

Wash

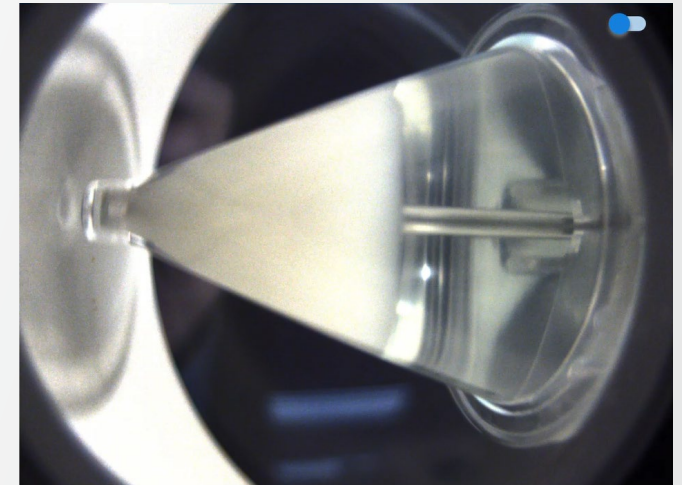
- Cells in the fluidized bed are rapidly exposed to new media resulting in very fast and efficient buffer exchange using a small volume of wash media.
- After creating a fluidized bed of cells in the original media, new wash media is pumped through the fluidized bed displacing the original media
- Typical washes are 3-5 chamber volumes (30-50ml)
- Washing with the same media, culture media (e.g. CTS™ AIM-V™ Medium), or PBS with 1–2% HSA, can significantly improve cell recovery.



0 mL



30mL



50mL

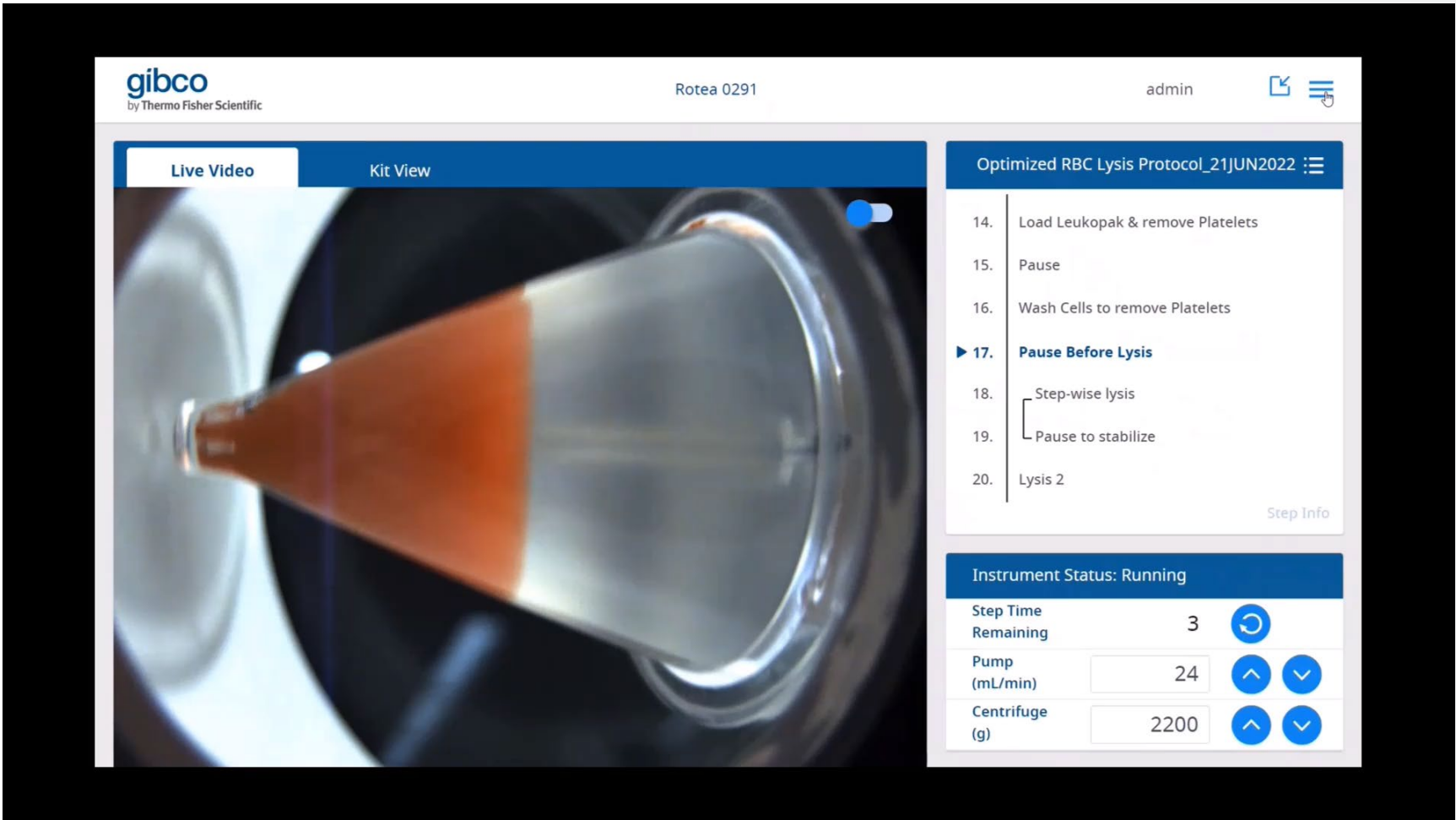
Buffer exchange

Differences in density/viscosity

- If the wash media is similar terms of density/viscosity to the original media, washing can often be conducted at the same settings as loading with minimal disruption of the fluidized bed.
- If the wash media is substantially different in terms of density/viscosity compared to the original media, the bed can be disrupted during the buffer exchange, which may cause cell loss.
 - Increasing the G force and/or reducing the flow rate can address this issue
 - Performing a step-wise wash may be optimal (add small amount of wash buffer, followed by 15s pause, repeat 1x or 2x, then follow up with single volume wash)
 - Elutriate or harvest cells to intermediate bag and dilute with wash media (1:5 to 1:10) and re-load mixture into cone

Buffer exchange

Differences in density/viscosity



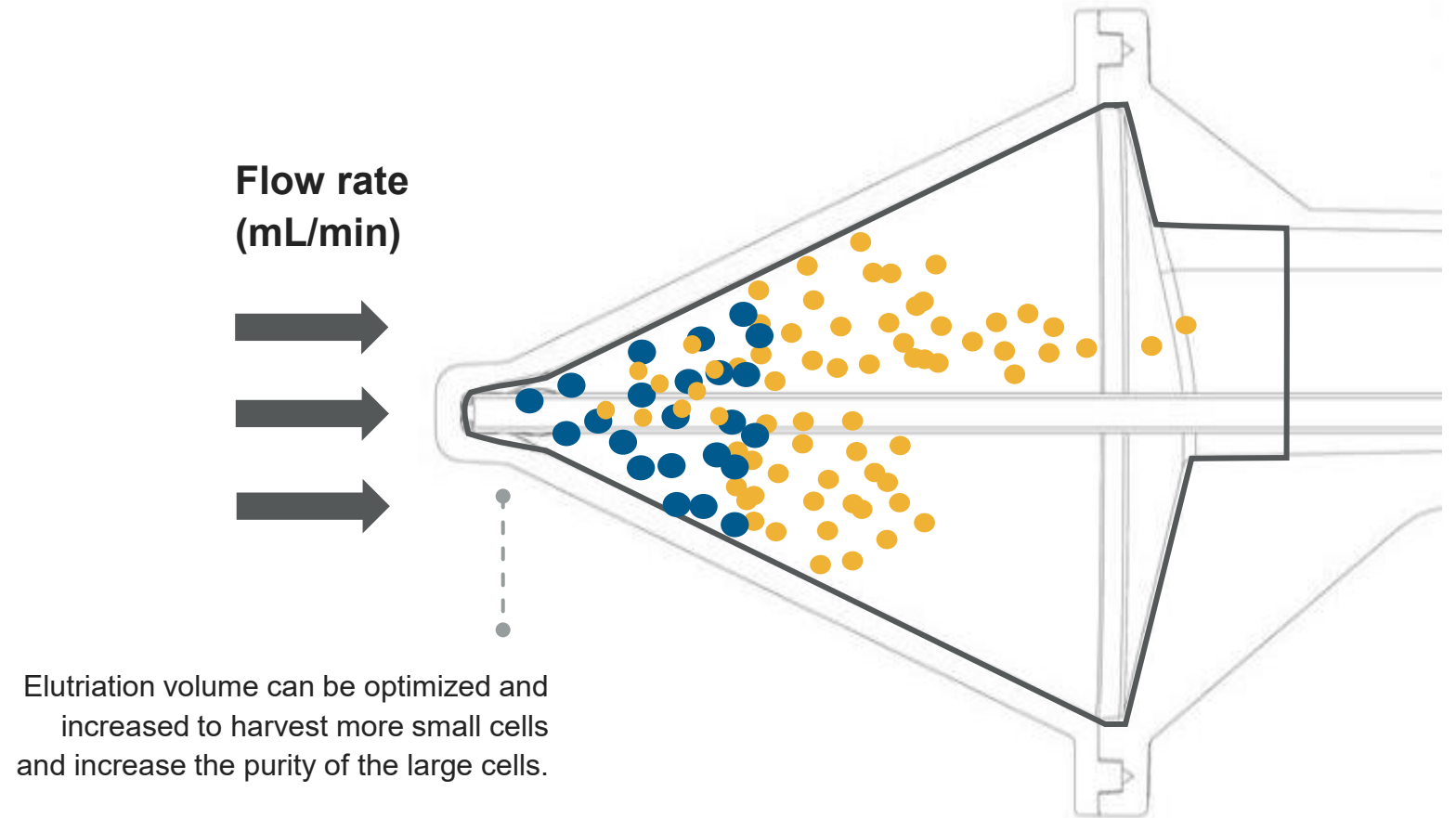
Elutriation

A process that enables particles of different size and/or density to be separated from each other

Flow rate and g-force settings can be adjusted to elutriate and separate smaller cells.

In this example, g-force is lowered. At the same flow rate, the larger cells stay within the chamber while small cells are elutriated.

When optimizing your settings, there is a balance between cell purity and cell recovery.



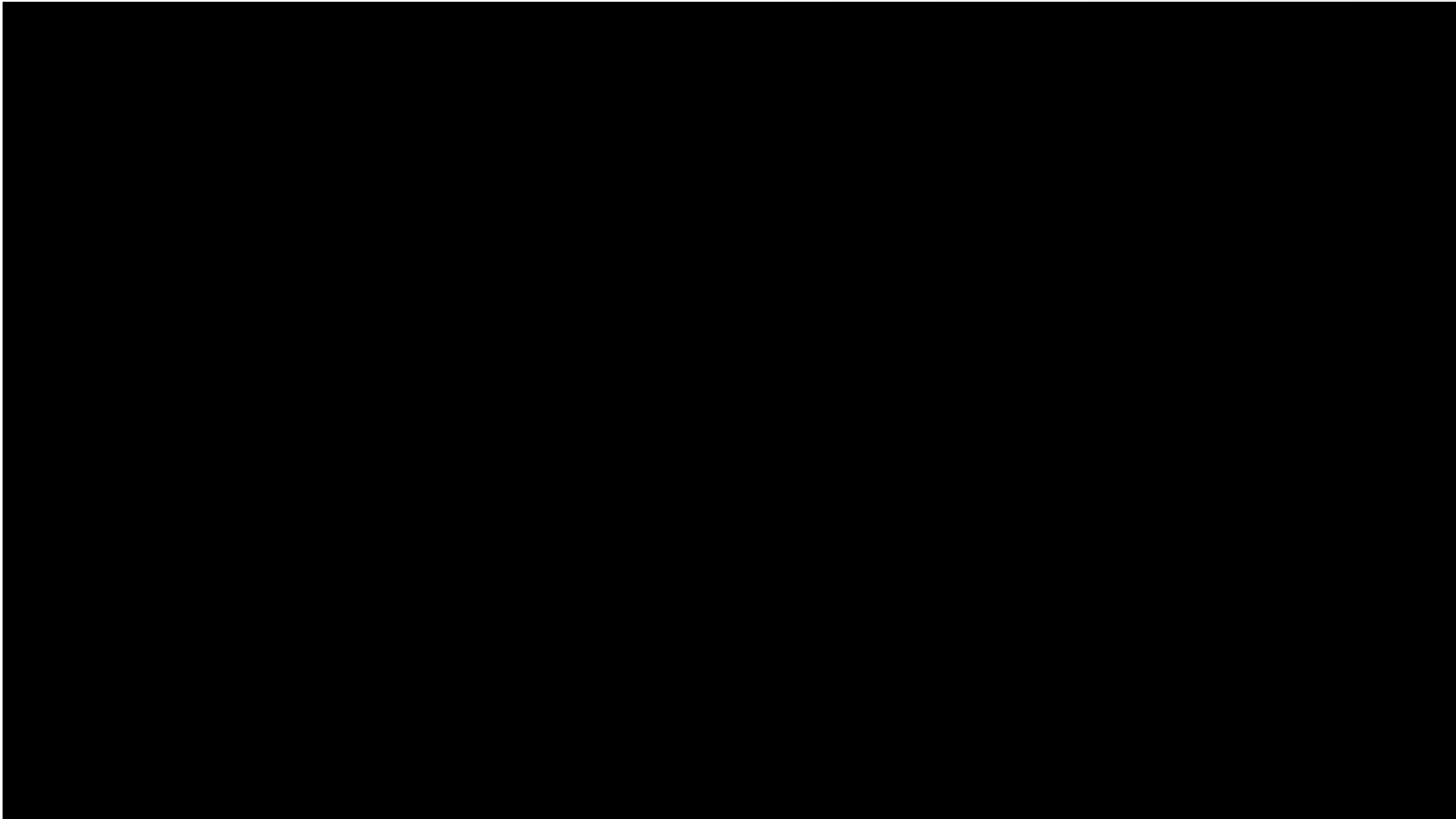
Elutriation



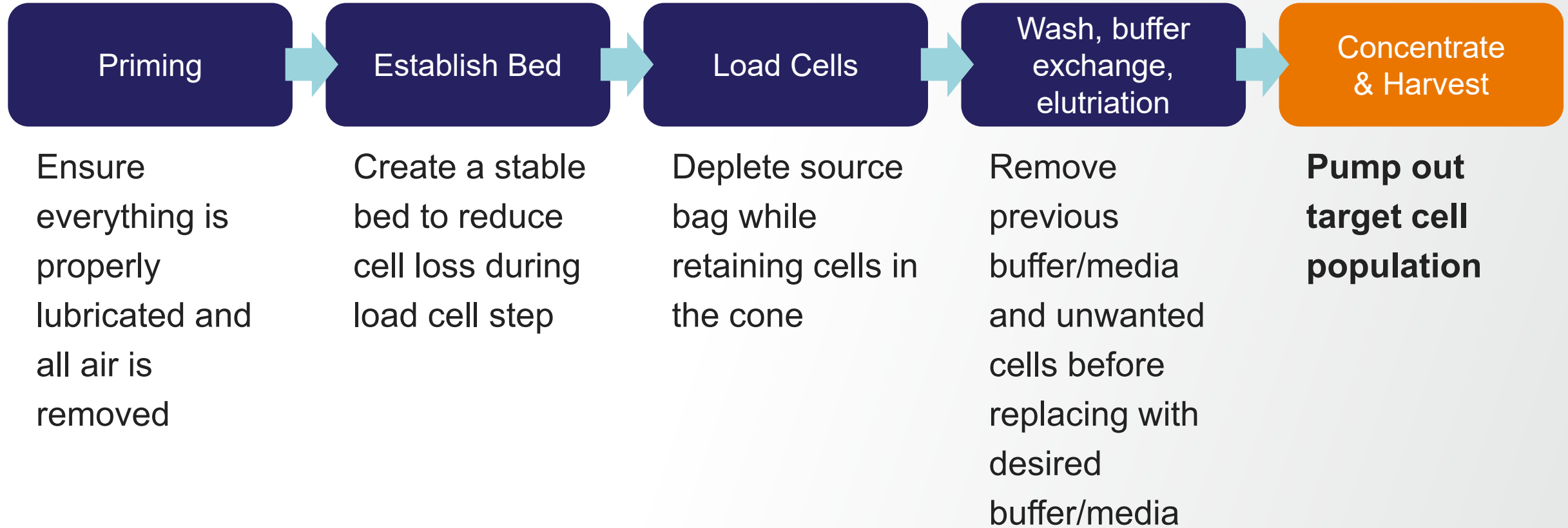
66 mL/min

68 mL/min

70 mL/min



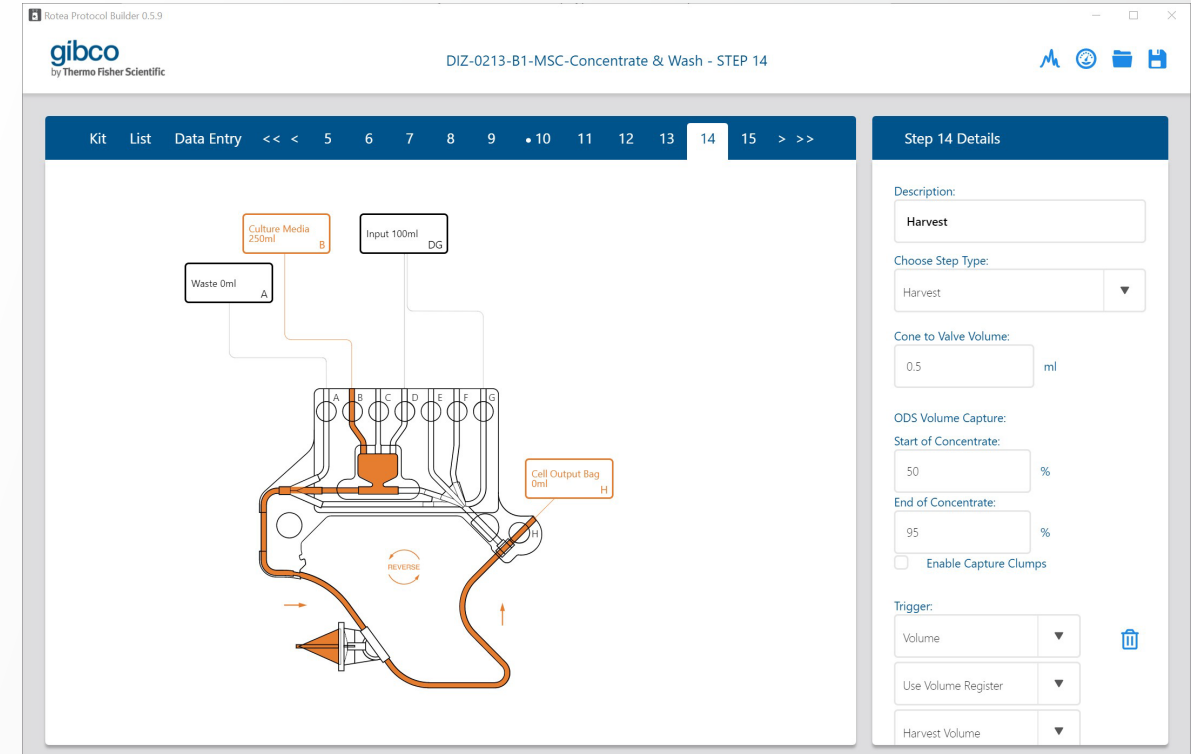
Basic Protocol steps



Concentrate & Harvest

Enables a high product concentration to be delivered by minimizing dead volume

- A brief (5-10s) pause step prior to harvest provides:
 - a step from which to restart a protocol after an interruption or stoppage
 - a smooth speed transition to concentrate the bed
- Harvest using line **E**, **F**, or **H**
- The Harvest step eliminates “dead volume” in the fluid path between the tip of the cone and the output valve.
 - Default is 0.5 mL, but it can be adjusted higher if desired ($\leq 1\text{mL}$)
- Minimum recommended harvest volume is 5mL, but higher cell numbers can benefit from $\geq 10\text{mL}$ harvest



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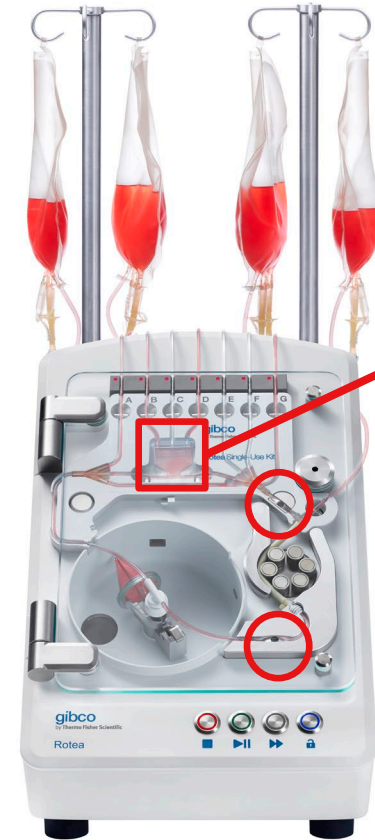
Troubleshooting

5

Technical Support

Blockage error

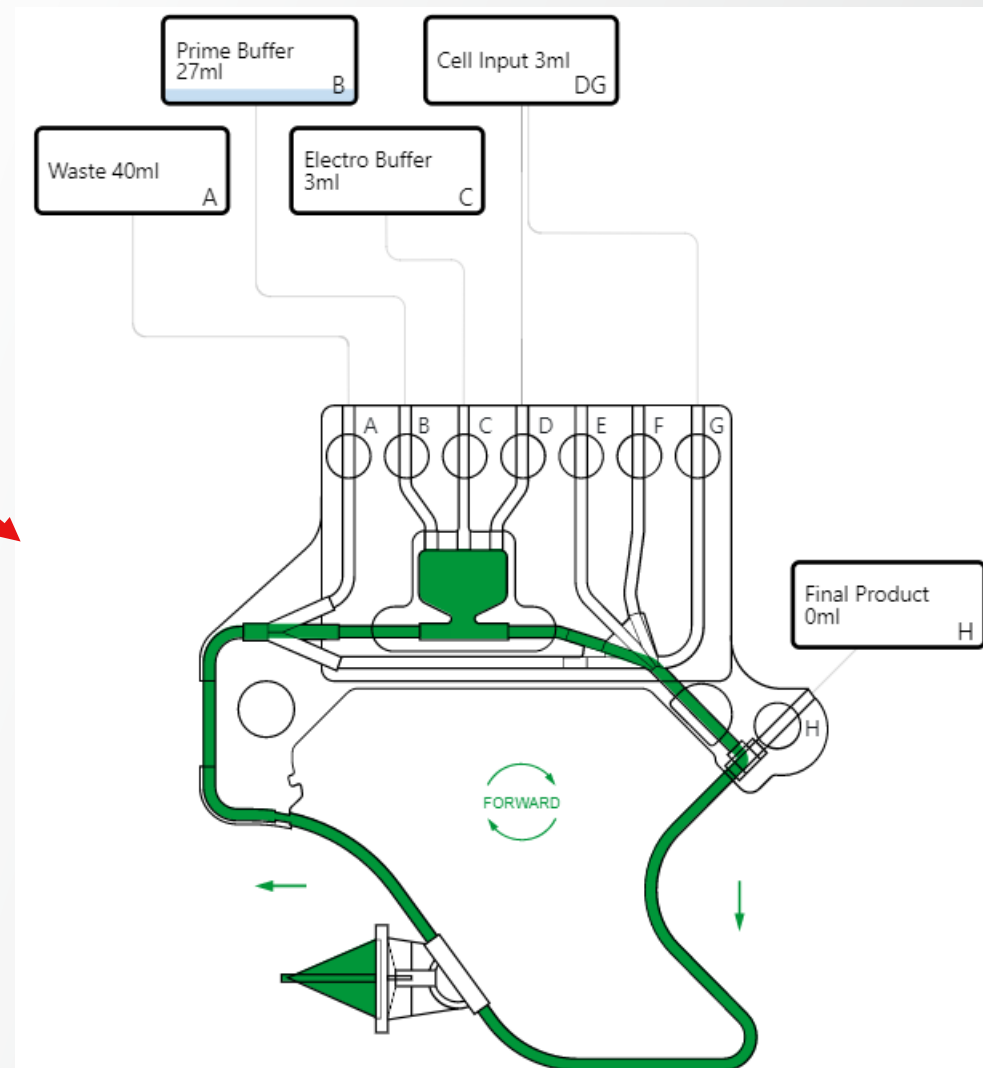
- Check lines for clamps, kinks, welds partially unopened
- If no obstructions, check for air at:
 - Pressure sensors
 - Bubble trap
- Air at pressure sensors appears as blockage
 - Pressure differential
- May require multiple restarts to re-prime
- Fill bubble trap step
- Clamp lines, open door, close door, restart from pause step



Mid-process issue

Use “Pause” instead of “Stop”

- If bed is unstable, cells elutriating, etc., use **“Pause”**
 - Allows cells/media to circulate
 - Protocol will still proceed
- If **“Stop”** is used, protocol ends
 - If protocol is stopped, recommend to resume protocol from a Pause step prior to stopped step
 - Any loop counts will be reset
- Intervention will be recorded in run log



Run Logs



C:\ProgramData\Rotea\run_logs

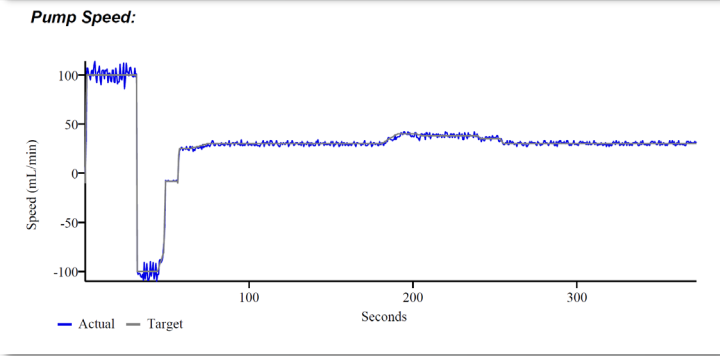
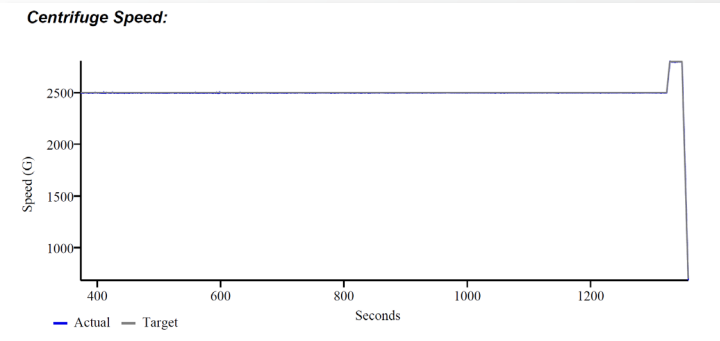
- Run log pdf not accessible while GUI is open
- The run logs contain:
 - instrument serial number, instrument software/firmware, time/date of run, kit info
 - protocol description, protocol steps, data entry values
 - volume that goes past valves, centrifuge speed, pump speed, ODS data
 - any errors that occurred

Protocol Run Parameters:

Parameter Name	Parameter Value	Units
Leukopak Dilution	30	ml
Intermediate Draw Volume	200	ml
Leukopak Draw Volume	65	ml
Intermediate Dilution	150	ml

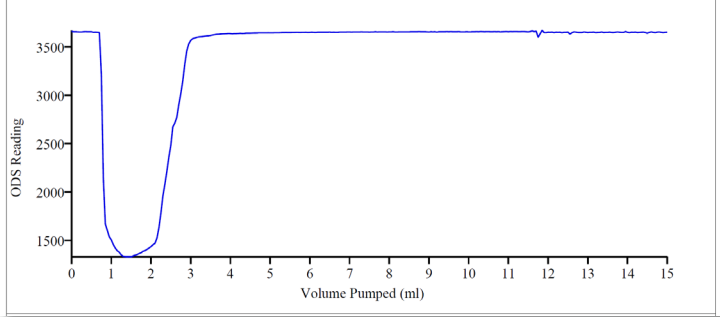
Run Log:

Time	Index	Description	Path	Step Type	Trigger	Centrifuge Speed (G)	Pump Speed (ml/min)	Run Time (s)
2022/06/22 16:18:38		User Login: admin, Level: admin						
2022/06/22 16:18:48	2	Lubricate Rotary Coupling	B to A	Normal	Volume	0	100	9
2022/06/22 16:18:58	3	Prime Chamber and Line A	B to A	Normal	Volume	10	100	9
2022/06/22 16:19:16	4	Add priming volume	B to A	Normal	Volume	10	100	18
2022/06/22 16:19:34	5	Add Leukopak Dilution Volume	B to A	Normal	Volume	10	100	18
2022/06/22 16:19:43	6	Prime Bubble Trap and Line B	A to B	Normal	Volume	10	-100	9



ODS Sample 1:

Step Number	DATE
Record Taken	
Minimum Threshold	1828
Maximum Threshold	3474
ODS Volume (ml)	2.17



Service Logs

C:\ProgramData\Rotea\service

- Service logs carry “raw” data/info from instrument
- Service logs can get **overwritten**
- Extract immediately after issue and share with Tech Support
- May need to upload to shared folder due to size

☐ Name	Date modified	Type	Size
 app_debug	11/1/2024 10:33 AM	Text Document	4,347 KB
 app_debug.log.1	9/25/2024 10:59 AM	1 File	10,240 KB
 app_debug.log.2	9/11/2024 3:10 PM	2 File	10,240 KB
 app_debug.log.3	9/11/2024 2:21 PM	3 File	10,240 KB
 app_debug.log.4	9/11/2024 10:57 AM	4 File	10,240 KB
 app_debug.log.5	9/10/2024 4:12 PM	5 File	10,240 KB
 app_debug.log.6	9/10/2024 3:36 PM	6 File	10,240 KB
 app_debug.log.7	9/10/2024 3:00 PM	7 File	10,241 KB
 app_debug.log.8	6/3/2024 3:13 PM	8 File	10,240 KB
 file1	10/9/2024 11:43 AM	Text Document	31 KB
 file2	10/9/2024 11:43 AM	Text Document	1 KB
 service	9/25/2024 11:13 AM	Text Document	7 KB
 splash_debug	11/1/2024 10:33 AM	Text Document	24 KB

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Q

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Cell & Gene Therapy > Cell Therapy Solutions > Cell Therapy Manufacturing > CTS Rotea Counterflow Centrifugation System

Resources

CTS Rotea Counterflow Centrifugation System
Resources

Get quote

Request a demo



Overview Features Specifications Accessories Software Sample Data Services Resources

Contact us Request demo

Documentation and literature

Demos and videos

Protocols

Software and firmware updates

Support

Learn more about cell therapy principles and applications

Explore additional resources for the Rotea system

Browse resources to find out more about the CTS Rotea system, including brochures, user guides, application notes, demos and videos, software, protocols, and education resources.

Have questions? See Rotea System product FAQs

CTS Rotea Counterflow
Centrifugation System Resources

Presentation Outline

1 Rotea Instrument and Single Use Kit

2 Introduction to Counterflow Centrifugation

3 Basic Protocol Steps

4 Troubleshooting

5 Technical Support

Instrument Service or Technical Support



Contact Options

 Call: 1-800-955-6288 opt. 2, 2, 1 for Technical Support; opt. 3, 1 for Instrument Service

 Email: instrumentservices@thermofisher.com Put “Rotea” in the subject line

 Visit: <https://www.thermofisher.com/us/en/home/technical-resources/contact-us.html>

For Instrument Issues:

Instrument Service

For instrument service, service contract, hardware support.

Rotea
Products
A47679 - CTS™ Rotea™ System + 2 year service
A47695 - CTS™ Rotea™ System + 2 year service + IQOQ
A49127 - Tube Clamps for CTS™ Rotea™ Single-Use Kit
A49585 - CTS™ Rotea™ Single-Use Kit
A46575 - CTS™ Rotea™ Hi-Flow Single-Use Kit

Enter the requested information in the following form.

Be sure to include:

- Instrument S/N
- Error message
- Software and firmware versions

For Technical Support or Kit Issues:

Technical Support

For technical support with product selection, application, troubleshooting.

Rotea
Products
A47679 - CTS™ Rotea™ System + 2 year service
A47695 - CTS™ Rotea™ System + 2 year service + IQOQ
A49127 - Tube Clamps for CTS™ Rotea™ Single-Use Kit
A49585 - CTS™ Rotea™ Single-Use Kit
A46575 - CTS™ Rotea™ Hi-Flow Single-Use Kit

To aid routing, select an Application Area *

-Select area-

-Select area-

Animal Health

Capillary Electrophoresis Sequencing & Fragment Analysis

Cell Analysis & Biochemical Assays

Cell Culture & Bioproduction

Cell Isolation & Expansion

Cloning & PCR

Diagnostics & Clinical Research

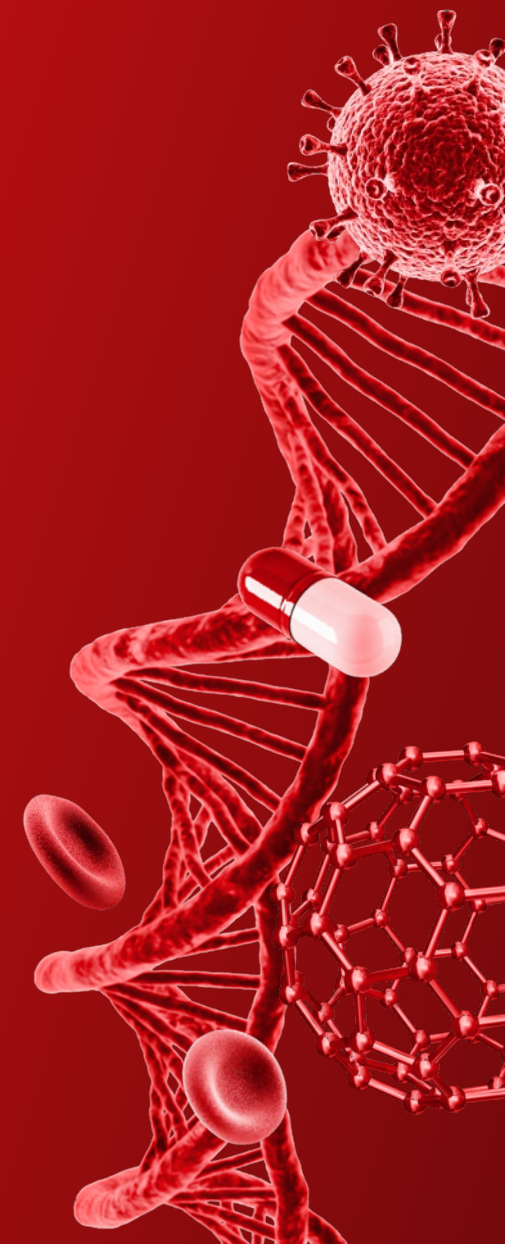
Choose the instrument or kit type and enter the requested information in the following form.

Choose “Cell Culture and Bioproduction” for the application area.

In the description field, please include the type of failure observed, the date of occurrence, and whether or not a patient batch was impacted.

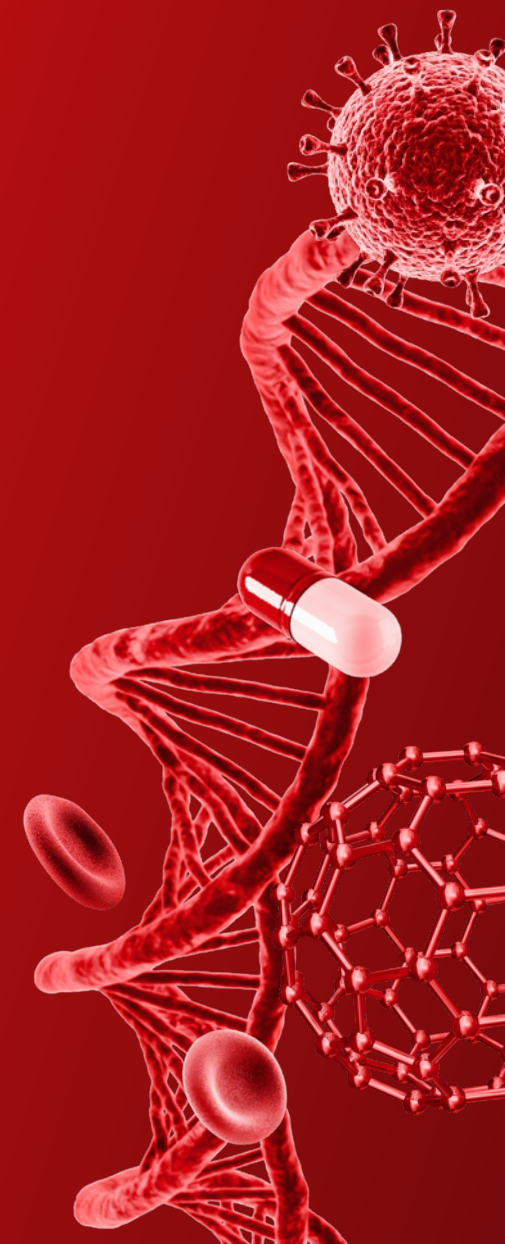
Thank you

Questions?



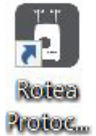
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Building a Protocol





- Rotea GUI
 - Control the device, user accounts, firmware
 - Only runs when a Rotea is connected to the PC



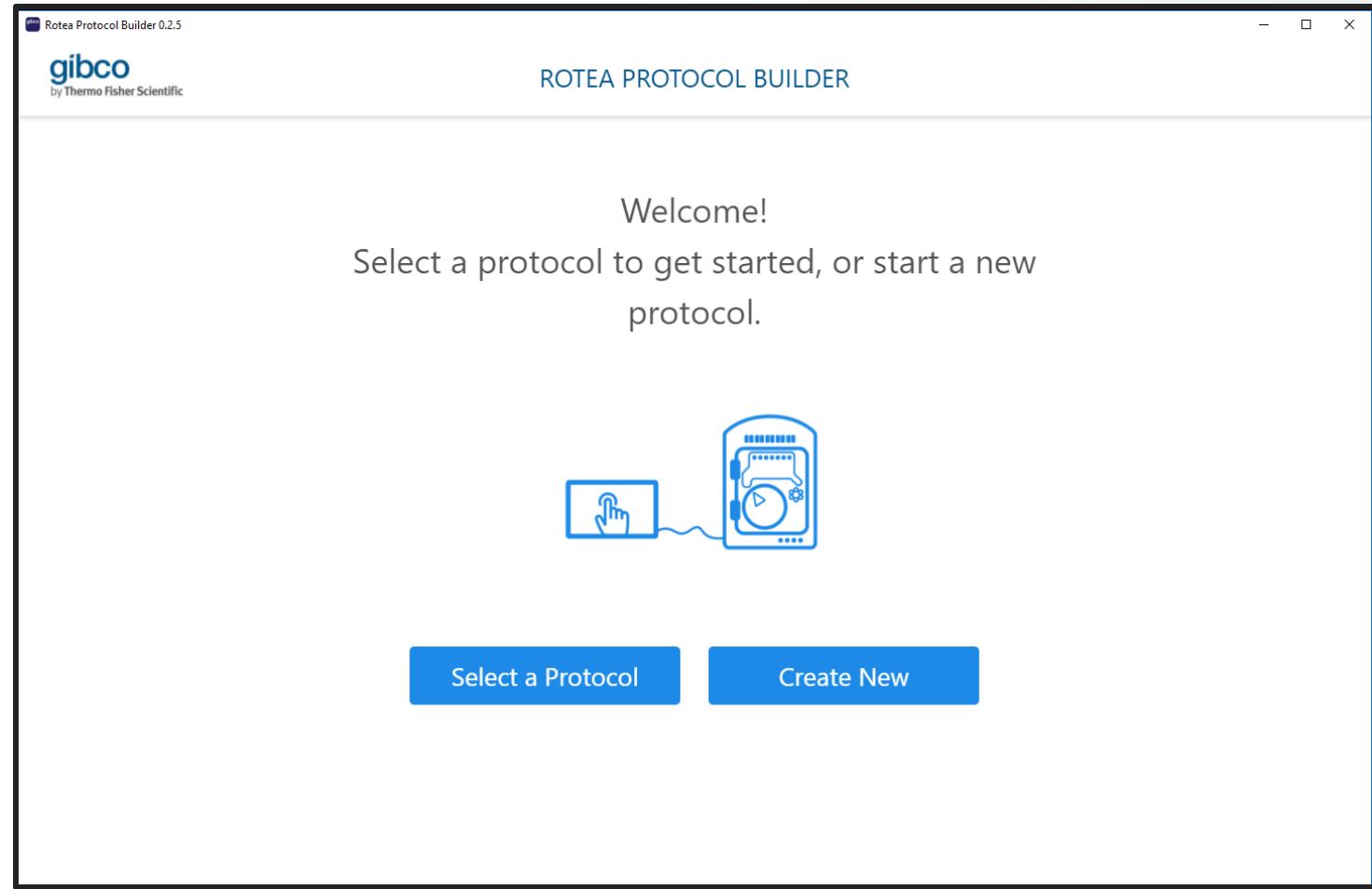
- Rotea Protocol Builder
 - Used for writing and modifying protocols
 - Can be installed and run on any Windows 10 PC

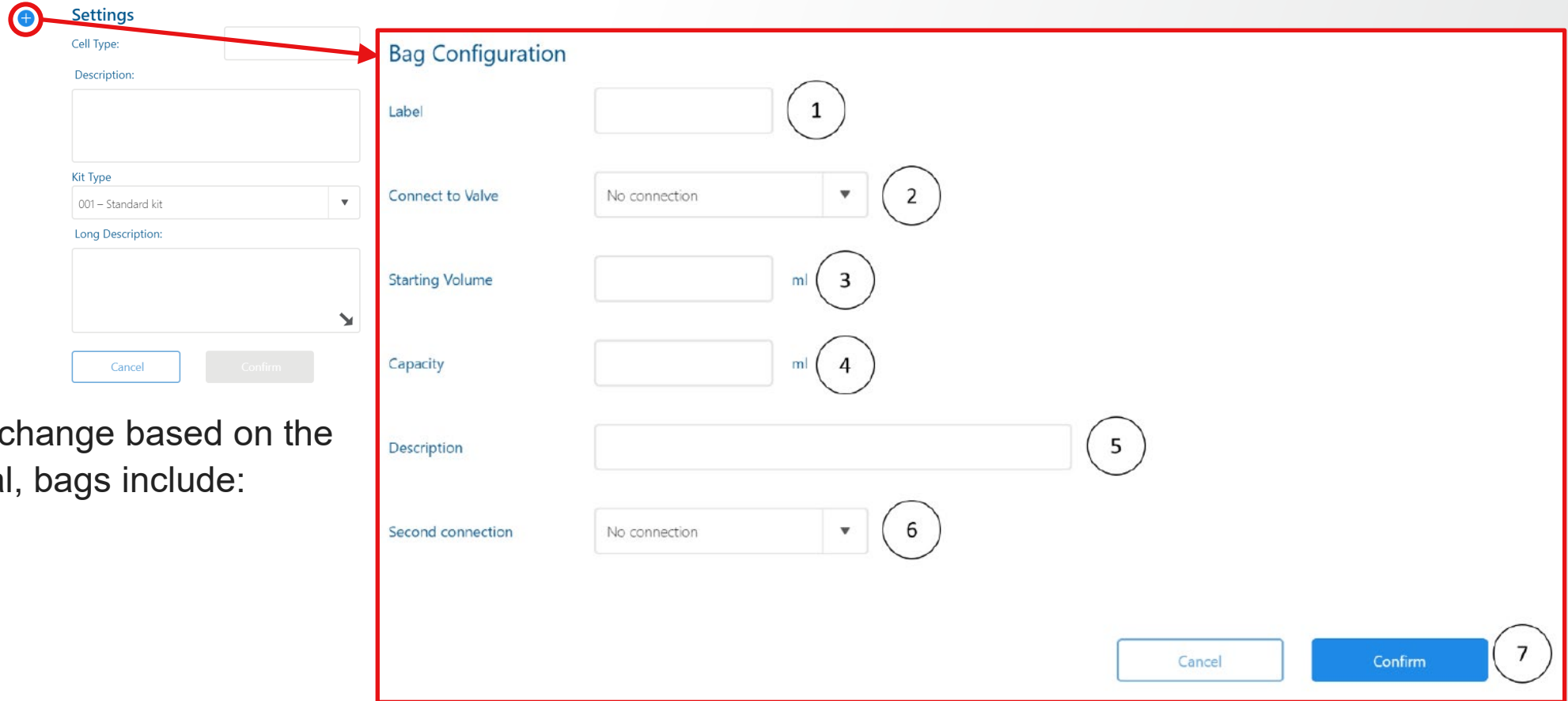
[Rotea Counterflow Centrifugation
System Software Download](#)



Rotea Protocol Builder

- Standalone app
- Creates .json files



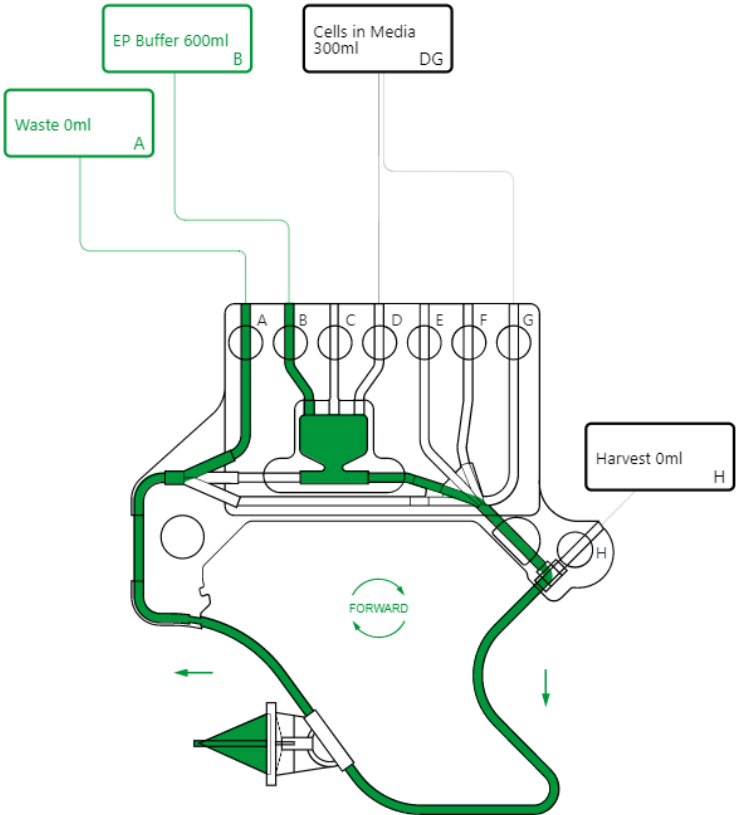


Note: Bags cannot be deleted if they are included in a protocol step

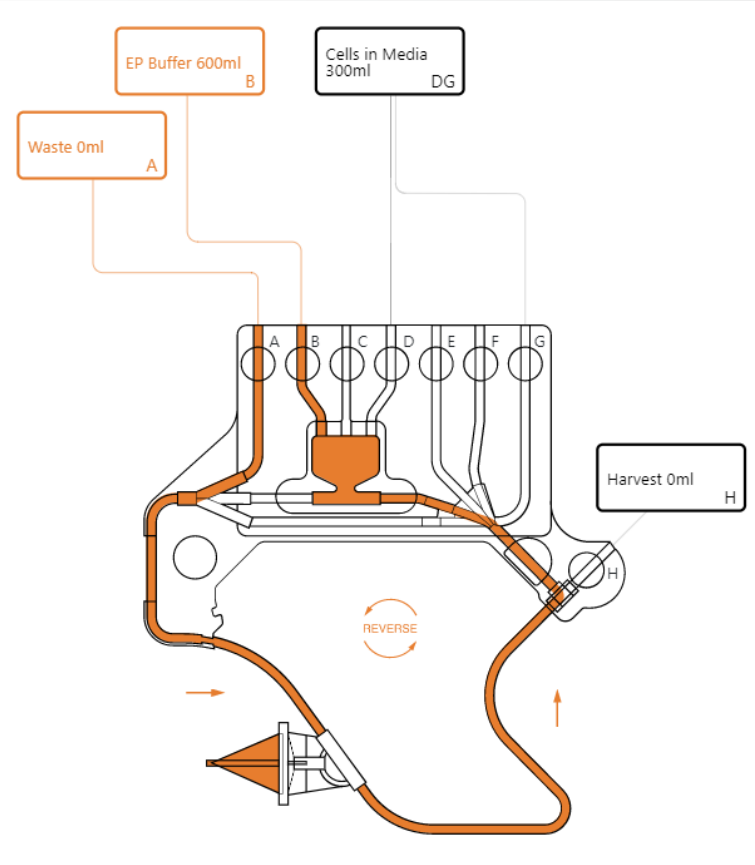
Adding Steps

Flow Path

Forward

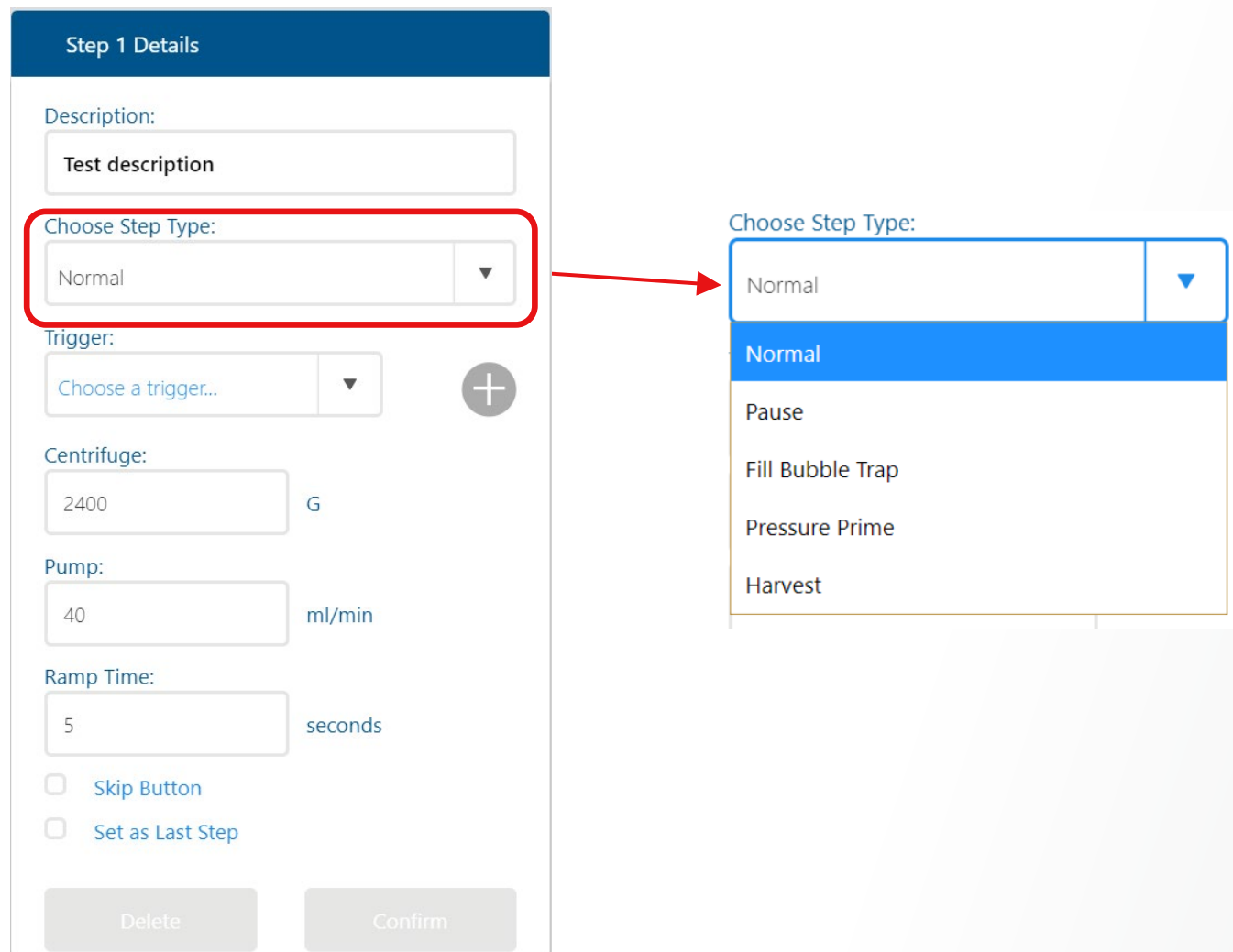


Reverse



Adding Steps

Step Type



Step 1 Details

Description:
Test description

Choose Step Type:
Normal ▼

Trigger:
Choose a trigger... ▼

Centrifuge:
2400 G

Pump:
40 ml/min

Ramp Time:
5 seconds

☐ Skip Button
☐ Set as Last Step

Delete Confirm

Choose Step Type:

- Normal
- Pause
- Fill Bubble Trap
- Pressure Prime
- Harvest

1. **Normal**— allows for volume to flow from one sample line to another
2. **Pause**— creates a closed loop fluid path within the kit that is isolated from the bags
3. **Fill Bubble Trap**— enables the bubble trap to be re-primed during a protocol without disrupting the fluidized bed. Only bags connected to lines B, C and D are available for selection for this step type
4. **Pressure Prime** — primes Lines E and/or F when they are not connected to a bag to minimize losses to these lines during a Harvest step
5. **Harvest**— is used to deliver a high concentration of cells to valves E, F or H with less carryover of the buffer.

Adding Steps

Triggers

Step 1 Details

Description:
Test description

Choose Step Type:
Normal

Trigger:
Choose a trigger...

Centrifuge:
2400 G

Pump:
40 ml/min

Ramp Time:
5 seconds

☐ Skip Button

☐ Set as Last Step

Delete

Confirm

Trigger:

Volume

Choose a trigger...

Volume

Timer

Pressure 1

Pressure 2

OD Sensor

Bubble Sensor Input

Bubble Sensor Output

☐ Pause on Bubble

The trigger is the condition that must be met before advancing to the next step

You can have multiple triggers

Step 13 Details

Description:
Elutriate Yellow Beads

Choose Step Type:
Normal

Trigger:
1 Volume

2 Use Volume Register

elutriateVolume

Multiplier:
1

Bubble Sensor Input

End of flow (Wet to Dry)

☐ Pause on Bubble

Adding Steps

Settings

Step 1 Details

Description:

Test description

Choose Step Type:

Normal

Trigger:

Choose a trigger...

Centrifuge:

2400

 G

Pump:

40

 ml/min

Ramp Time:

5

 seconds

☐ Skip Button

☐ Set as Last Step

Delete

Confirm

Enter the desired G force

Enter the pump speed in mL/min

Ramp time = time course for applying changes in G force and pump speed based on previous step. Longer ramp times may be desirable to apply more gradual changes



Skip Button



Set as Last Step

List View



Kit

List

Data Entry

<< <

1

2

3

4

5

6

7

8

9

10

• 11

> >>

Summary: Basic wash and concentrate protocol

Step	Description	Flow Path	Speed	Flow Rate	Step Type	Triggers
1	Pre-prime	B to A	0g	100ml/min	Normal	Input Bubble Sensor
2	Lubricate Rotary Coupling	B to A	0g	100ml/min	Normal	Volume: 15 ml
3	Prime Chamber and line A	B to A	10g	100ml/min	Normal	Volume: 15 ml
4	Add priming volume	B to A	10g	100ml/min	Normal	Volume: 20 ml
5	Prime Bubble Trap and line B	A to B	10g	100ml/min	Normal	Volume: 15 ml
6	Prime line D	A to D	10g	50ml/min	Normal	Volume: 5 ml

Copy To

Move To

Insert

Repeat

Add Label

Delete

Insert Step

Before / After

After

Before

After

1

1

2

3

4

5

6

7

Cancel

Confirm

Loops



Kit

List

Data Entry

<< <

1

2

3

4

5

6

7

8

9

10

• 11

> >>

Summary: Basic wash and concentrate protocol

	Step	Description	Flow Path	Speed	Flow Rate	Step Type	Triggers
☒	9	Ramp speed to load	J to K	2500g	30ml/min	Pause	Time: 15 seconds
	10	Establish bed	D to G	2500g	30ml/min	Normal	Volume: 1 x Establish Bed Volume ml
	• 11	Load Input material	D to A	2500g	40ml/min	Normal	Input Bubble Sensor, pause
	12	Wash	B to A	2500g	40ml/min	Normal	Volume: 50.0 ml
	13	Concentrate bed for Harvest	J to K	2800g	20ml/min	Pause	Time: 10 seconds
☒	14	Harvest	B to H	2800g	50ml/min	Harvest	Volume: 1 x Harvest Volume ml
	15	ramp to stop	J to K	700g	20ml/min	Pause	Time: 10 seconds

Copy To

Move To

Insert

Repeat

Add Label

Delete

Loops

Step	Description
8	Establish bed
9	Load cells
10	Concentrate bed for wash
11	Wash
12	Concentrate bed
13	Harvest

14 Ramp to stop

You then select which condition must be met before advancing to the steps listed after the loop.

Do Steps While...

Condition

No selection

Less than

Value

OR

No selection

Cancel

Confirm

No selection

- Repeat Count (number of times to repeat)
- Bubble Sensor B Not Triggered
- Bubble Sensor D Not Triggered
- Process Counter 1
- Process Counter 2
- Process Counter 3
- Process Counter 4
- Timer (Seconds)
- Timer (Minutes)
- Skip Button Not Triggered

Data Entry

The screenshot displays the 'Data Entry' tab in a software interface. At the top, there are tabs for 'Kit', 'List', and 'Data Entry', with 'Data Entry' being the active tab. Below this, there are two data entry fields. The first field is named 'Draw volume' with units 'mL', a minimum value of '0.00', and a maximum value of '1000.00'. The second field is named 'Harvest volume' with units 'mL', a minimum value of '0.00', and a maximum value of '1000'. Each field has a trash icon and a plus icon. Below the fields, there are three panels showing the configuration options for the 'Draw volume' field. The first panel shows the 'Trigger' dropdown set to 'Volume' and the 'Value' field set to '100 ml'. The second panel shows the 'Trigger' dropdown set to 'Use Volume Register' and the 'Multiplier' field set to '1'. The third panel shows the 'Trigger' dropdown set to 'Choose a register...' and the 'Register' dropdown set to 'Draw volume'. Red arrows indicate the flow from the 'Target Volume' field in the first panel to the 'Use Volume Register' option in the second panel, and then to the 'Draw volume' option in the third panel.

Kit List Data Entry 1 2

Name: Draw volume Units: mL Min: 0.00 Max: 1000.00

Name: Harvest volume Units: mL Min: 0.00 Max: 1000

Trigger: Volume

Target Volume

100 ml

Trigger: Volume

Use Volume Register

Choose a type...

Target Volume

Use Volume Register

Multiplier: 1

Trigger: Volume

Use Volume Register

Choose a register...

Choose a register...

OD Sensor

Valve A

Valve H

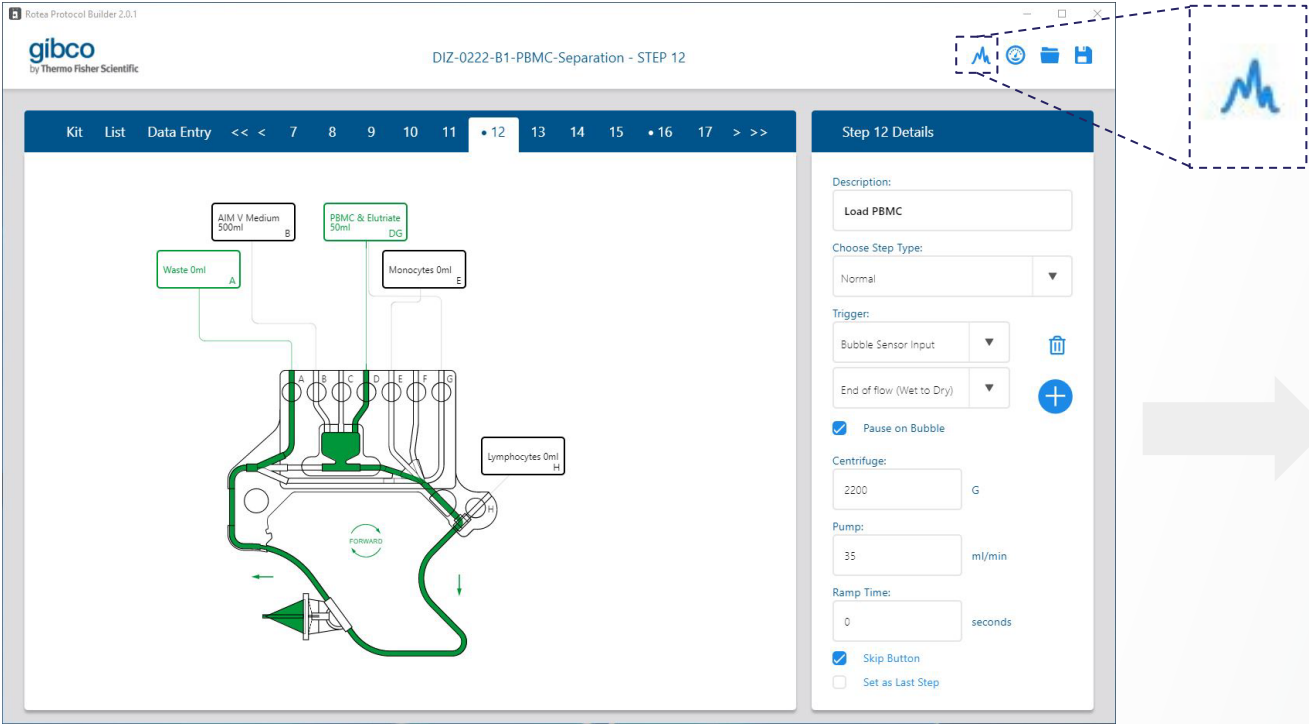
Draw volume

Harvest volume

The Data Entry screen enables the creation of variable volumes

- Useful for conditions that depend upon sample, such as establish bed volume (since the volume for 50 million cells will likely change with each sample/run)
- A Data Entry field can't be deleted once it's used in a protocol – it can be renamed
- Can't be used for flow rates or G force

Process Modeling



Settings

Model: Loading Leukocytes

Media: AIM V Medium 100%

Density: 1008 kg/m³

Viscosity: 1.1 kg/ms (x10⁻³)

Speed:

Centrifuge: 0 G

Flow rate: 0 ml/min

G:F Ratio: Enter Ratio

Cancel Save & Exit

1. Assign a model name
2. Select media flow through chamber in that step. If mixture of media are used, 2 media can be selected. Try to best estimate their ratios.

Process Modeling

3. Choose the cell type from the list

OR

Create a new particle if one is not present in the list

Up to 4 particles can be added for comparison

Rotea Protocol Builder 2.0.1

gibco
by Thermo Fisher Scientific

DIZ-0222-B1-PBMC-Separation - PROCESS MODEL

Add Particle

Particle: **3.** Please Choose ▼

Cell Density: 0 kg/m3

Cell Diameter: 0 microns

Compensation Factor:

3a. Create New

Cancel Add

G:F Ratio: Enter Ratio 🔒

Cancel Save & Exit

Cells retained Cells elutriated

The use of this Rotea™ Simulation Software and Rotea™ Process Model is for informational purposes only and is not intended to substitute empirical data.

Process Modeling

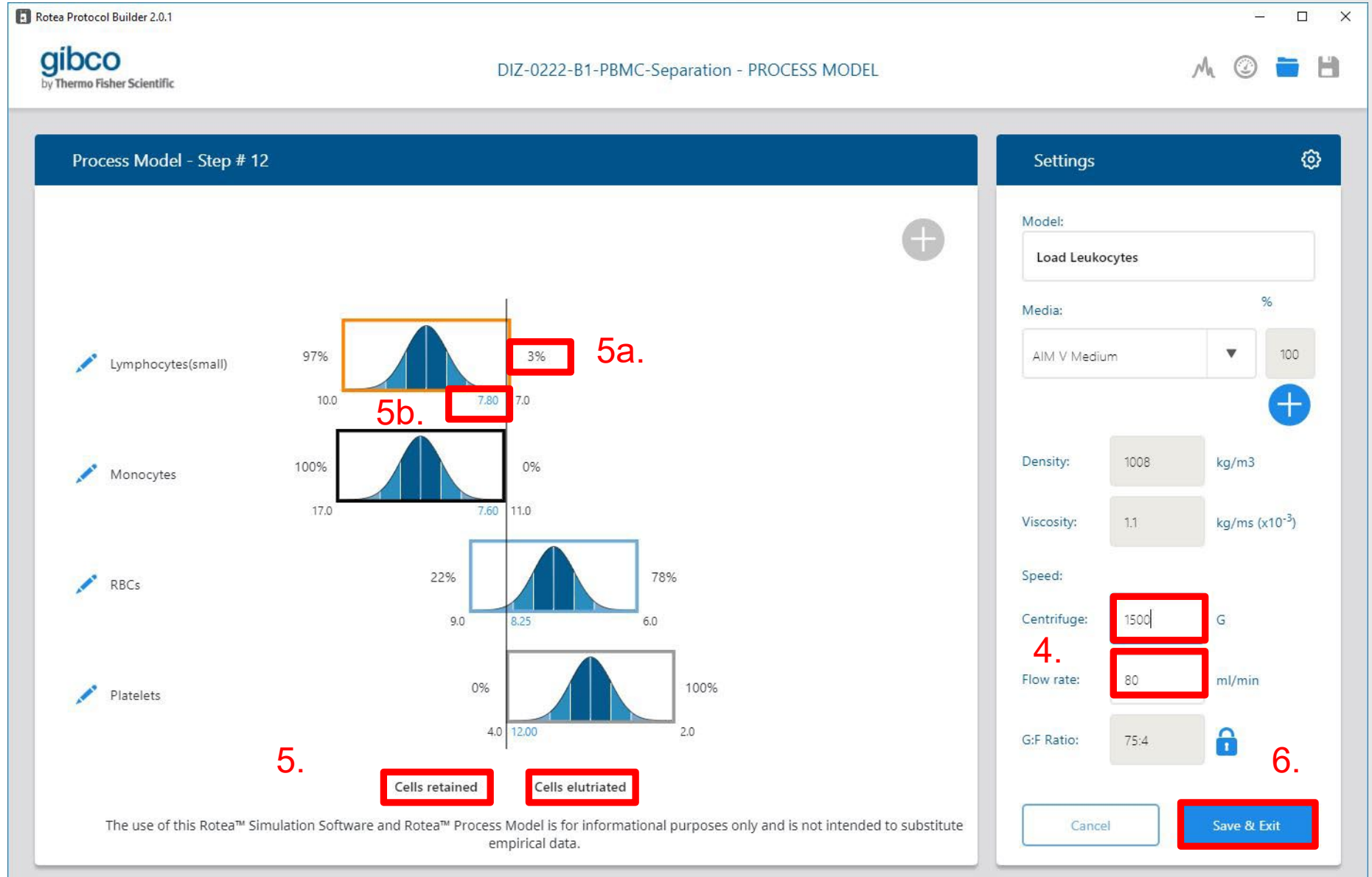
4. Change the centrifuge and flow rate values to see what cells will retain or elutriate

5. Left side shows cells retained and right is cells elutriated

a) Percentage of cells in each category

b) Size of cells (μm) that will retain versus elutriate

6. Save and Exit



6.

4.

5a.

5b.

5.

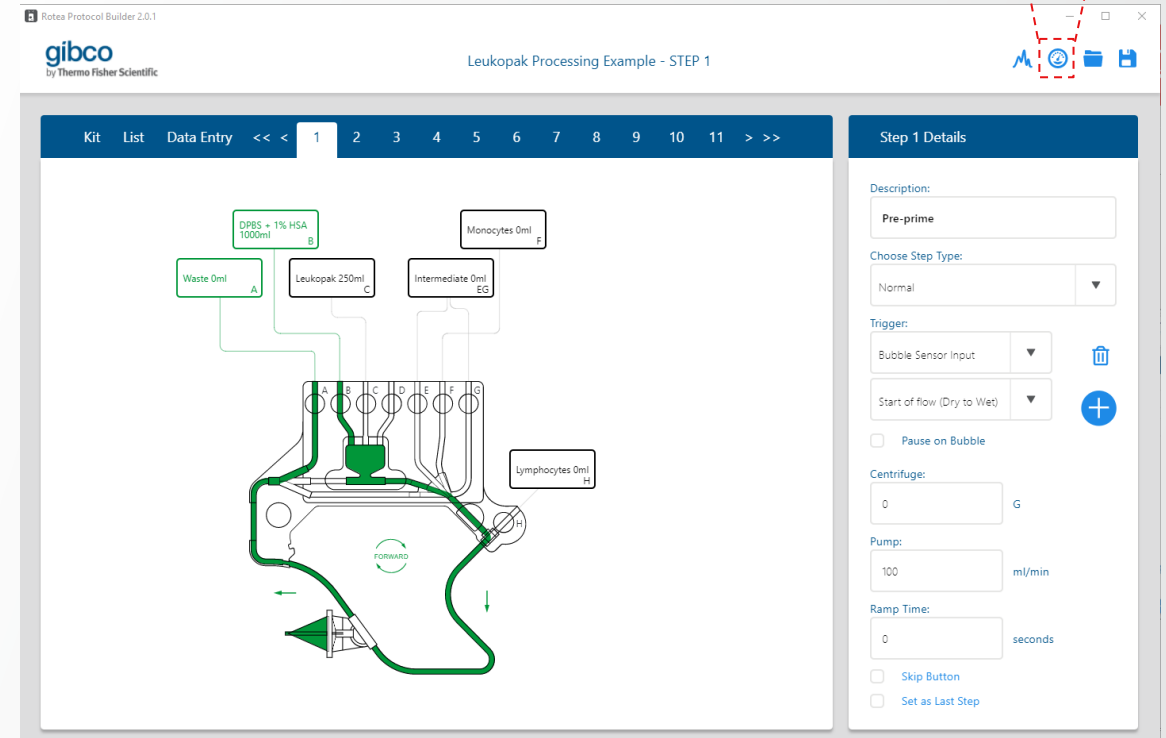
Protocol Simulator

Allows a protocol to be simulated by the protocol builder software

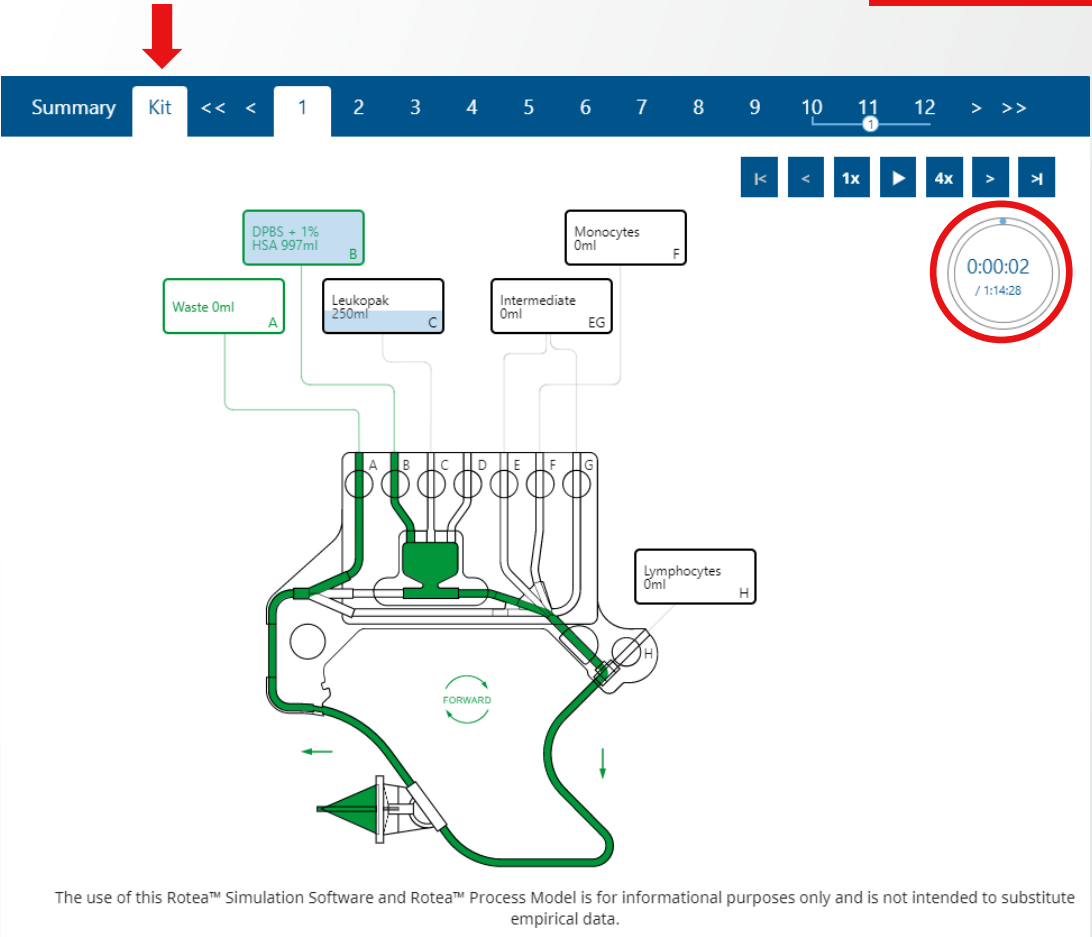
Provides an estimate of:

- Process Duration
- Volume Consumption
- Waste Generation

Allows operator to confirm steps execute as intended



Protocol Simulator



Process volumes and capacities are entered into the “Define Kit” tab in the Protocol Builder Software

Data Entry values entered for a specific run

Data entry:

Leukopak Dilution	250.00 ml	Leukopak Aliquot	125.00 ml
Harvest Volume	25.00 ml		

Cancel Confirm

Process simulation controls and process duration estimate available from the “Kit” tab

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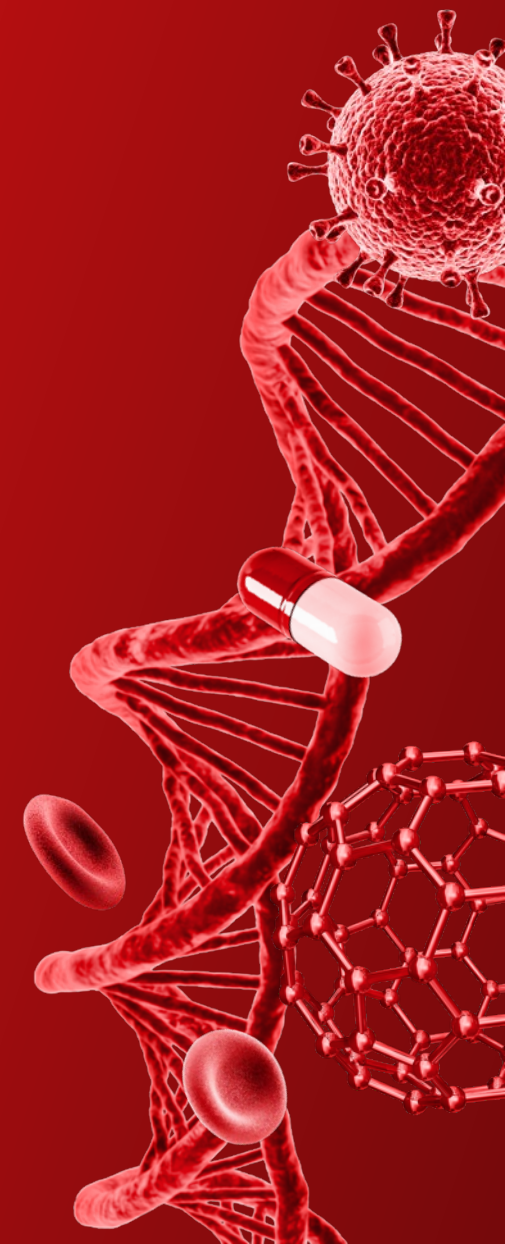
B – Indicates the protocol will run out of buffer before completion based on the values input in the “Define Kit” tab

1129.5 B

1177.0 A

1000 A

Supplemental Content



Governing Equations – Force Balance

Duplicate slide

$$F_{\text{net}} = F_{\text{Centripetal}} - F_{\text{Stokes Drag}}$$

$$F_{\text{Centripetal}} = \frac{(\rho_p - \rho_m) \pi d^3 \omega^2 r}{6}$$

ρ_p = Density of particle

ρ_m = Density of media

d = Diameter of particle

ω = Angular velocity

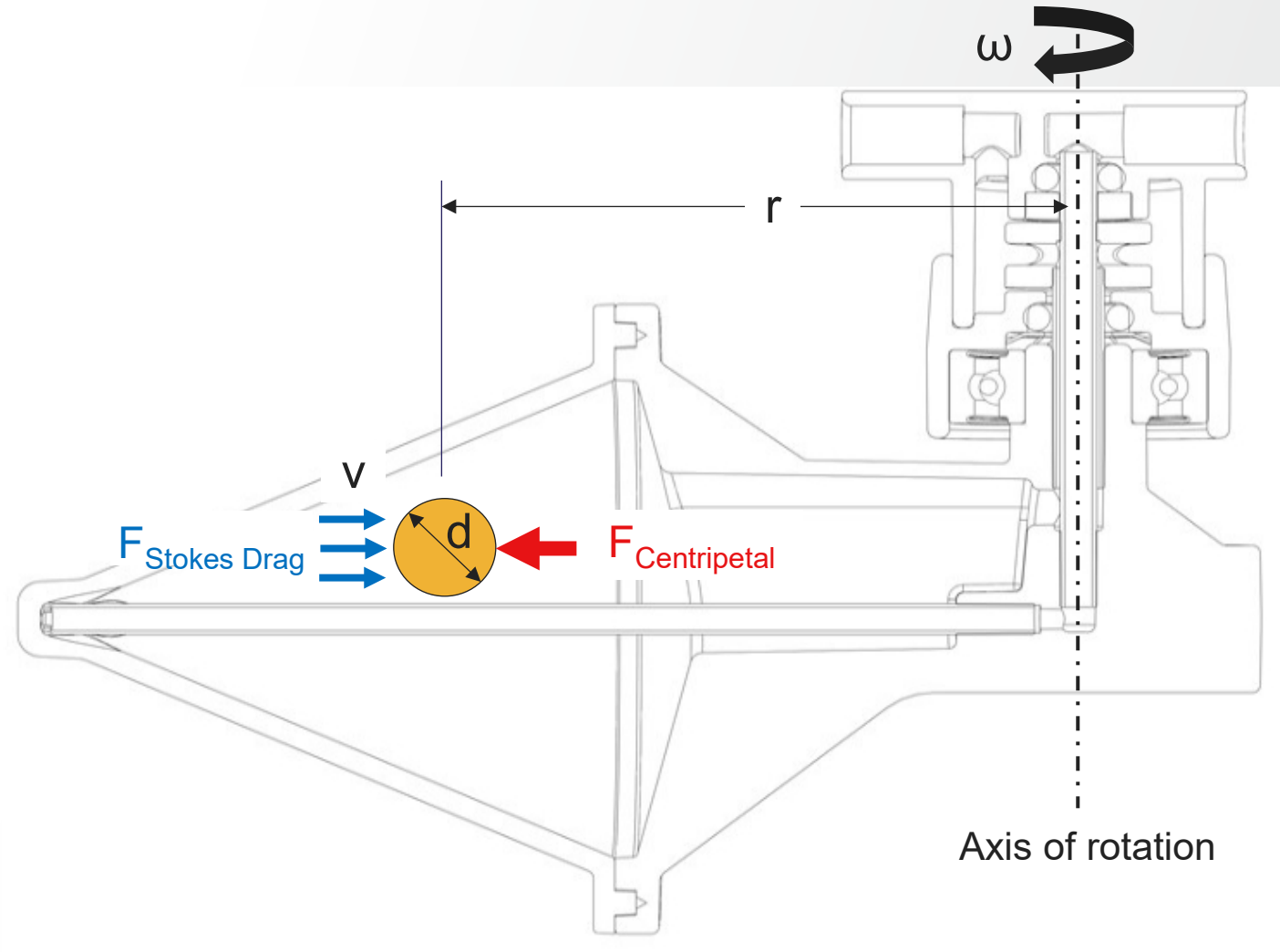
r = Radial position of particle

$$F_{\text{Stokes Drag}} = 3\pi\eta vd$$

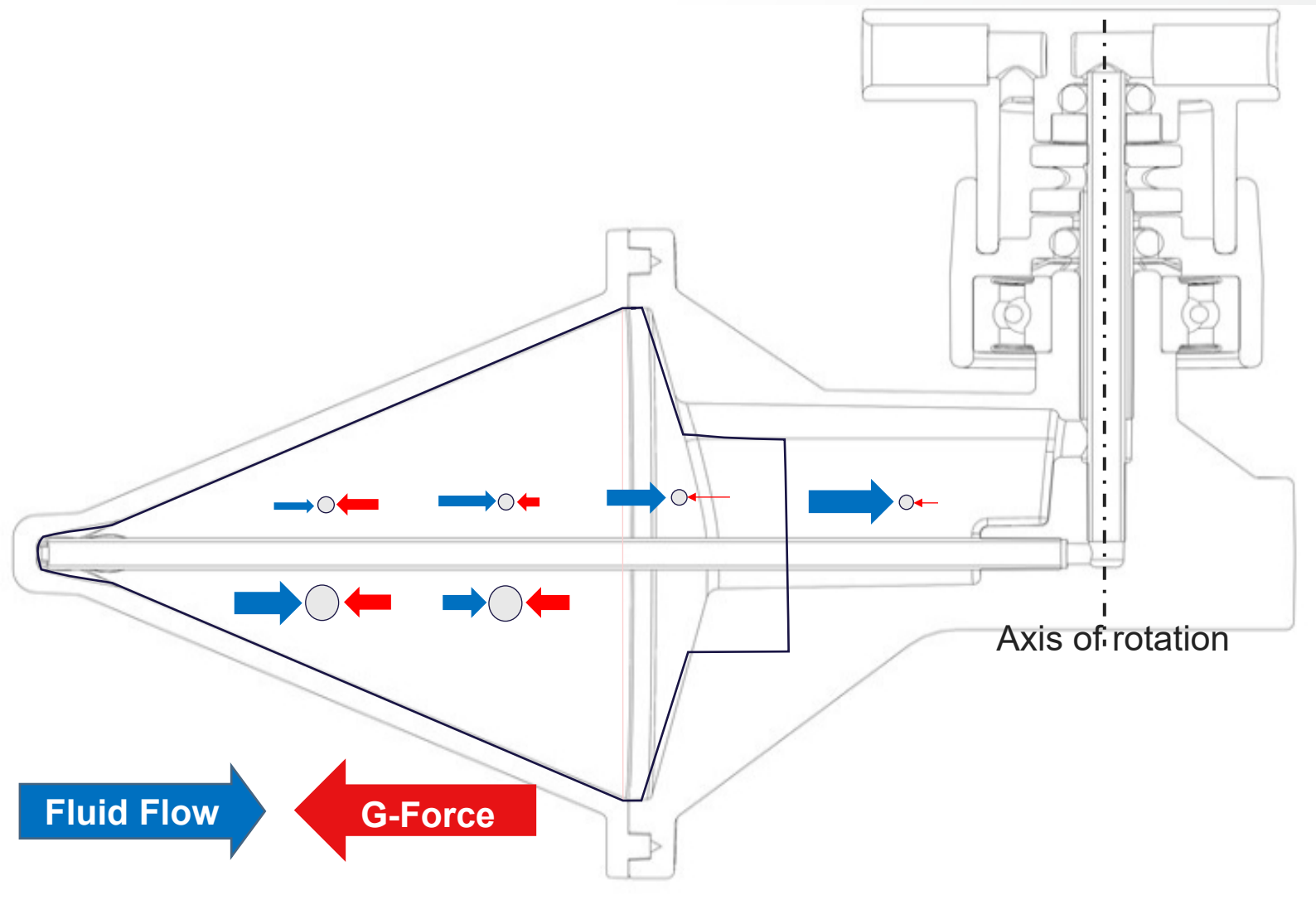
η = Viscosity of media

v = Velocity of fluid at particle location

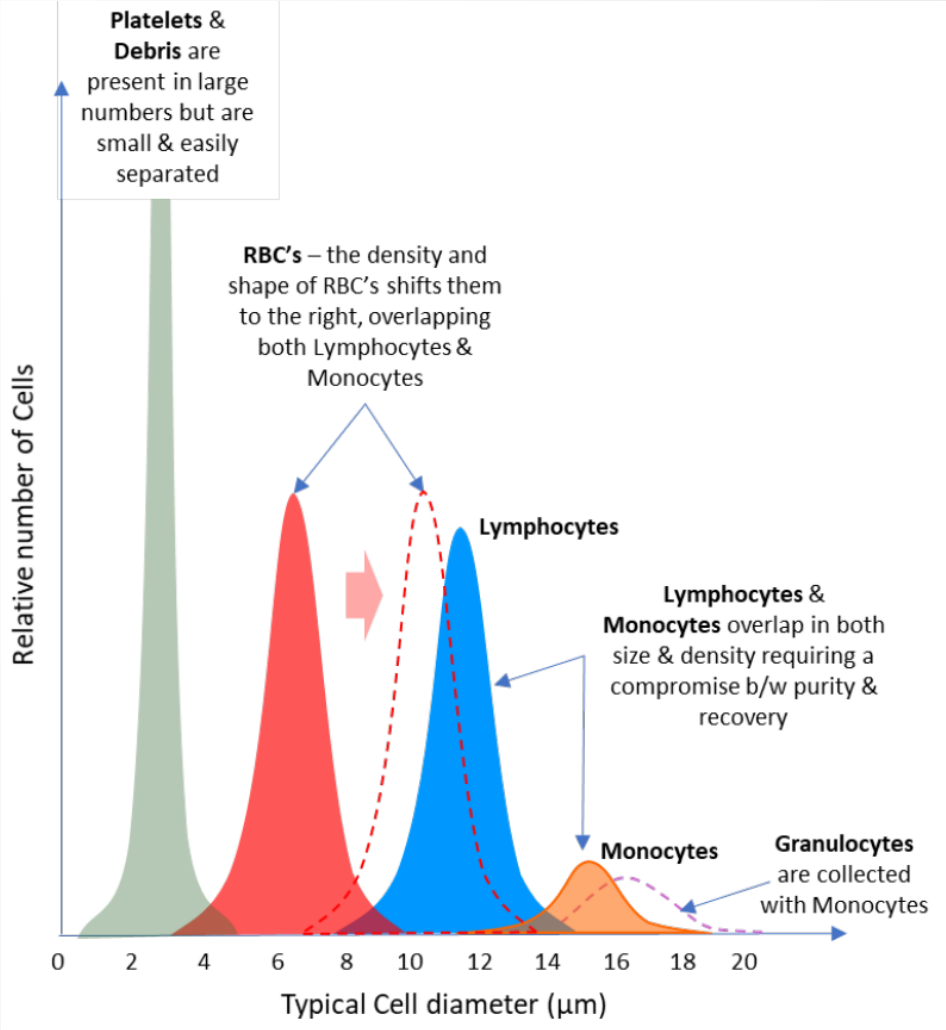
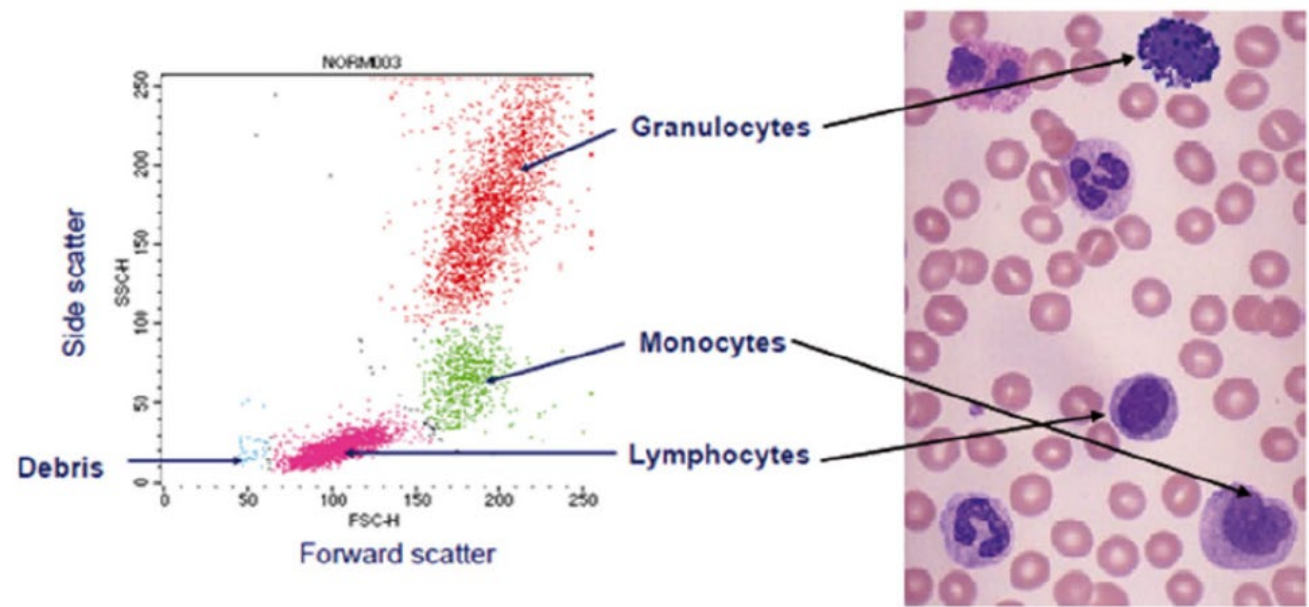
d = Diameter of particle



Introduction to Counterflow Centrifugation



Leukopak Processing



Special Protocol Steps

RBC lysis

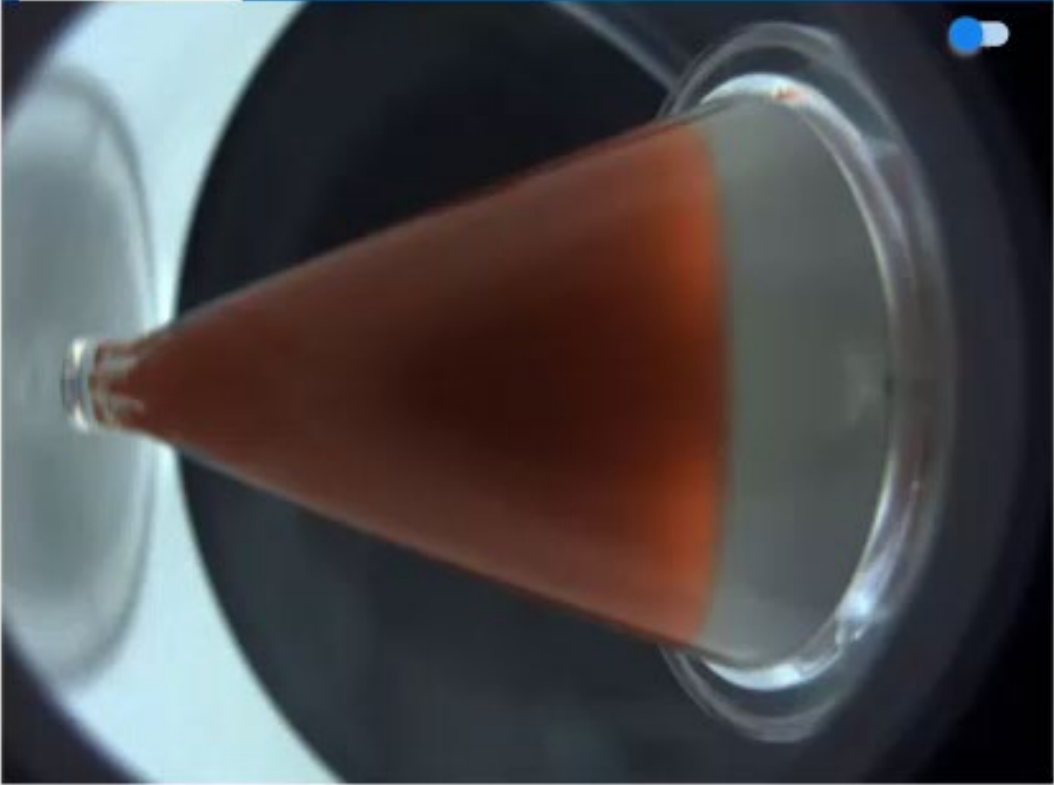
Leukopak Run 2
~2.7 billion cells in 42ml
106ml of ACK lysis used
Harvest PBMCs

Special Protocol Steps



Monocyte and Lymphocyte separation

Live VideoKit View



Leukopak Processing

10. Adjust Speeds For Loading

11. Load Leukopak

12. Pause Before Wash

13. Wash Cells

14. Pause

▶ 15. Adjust Speeds to Elutriate Lymphocytes & ...

16. Elutriate Lymphocytes & RBCs

17. Pause

18. Concentrate Monocytes

19. Harvest Monocytes

20. ...

Step Info

Instrument Status: Running

Step Time Remaining9

Pump (mL/min)28

Centrifuge (rpm)1314

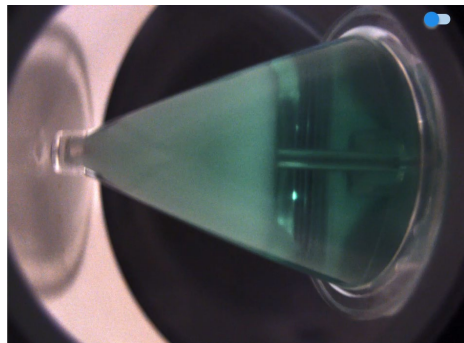
↺

⬆️⬆️

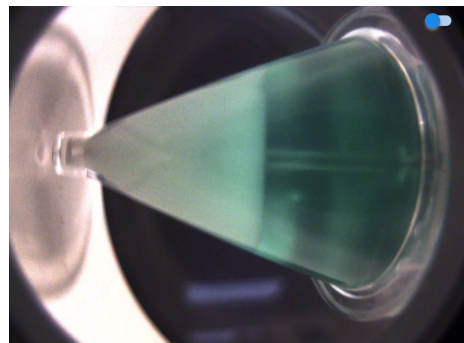
⬆️⬆️

Visualization of Residual Washout

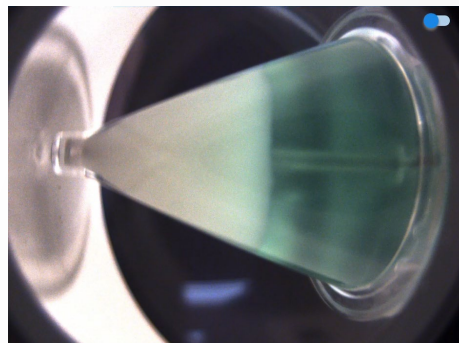
0mL Wash



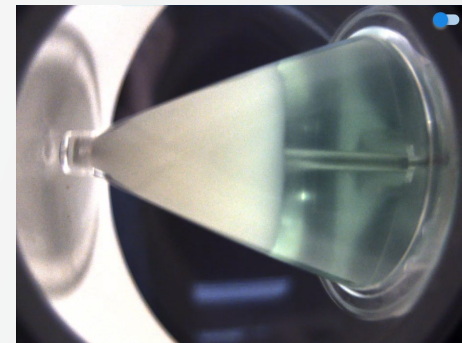
10mL Wash



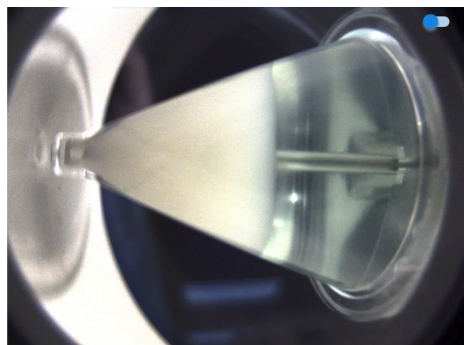
20mL Wash



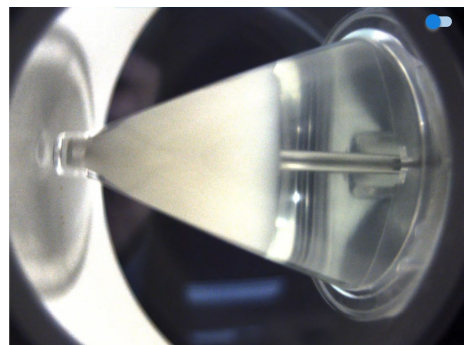
30mL Wash



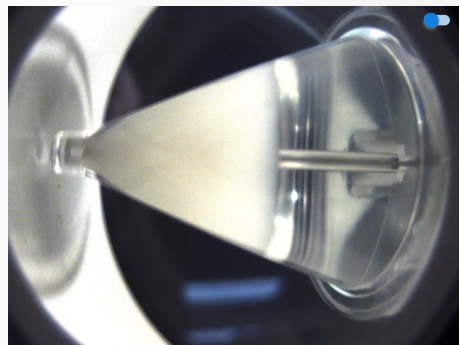
40mL Wash



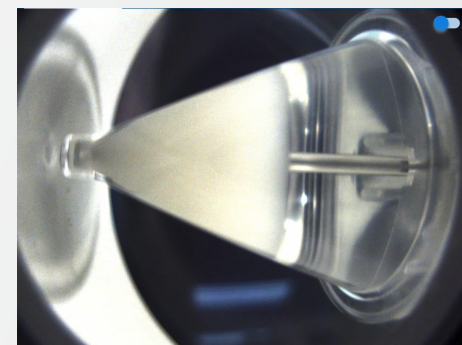
50mL Wash



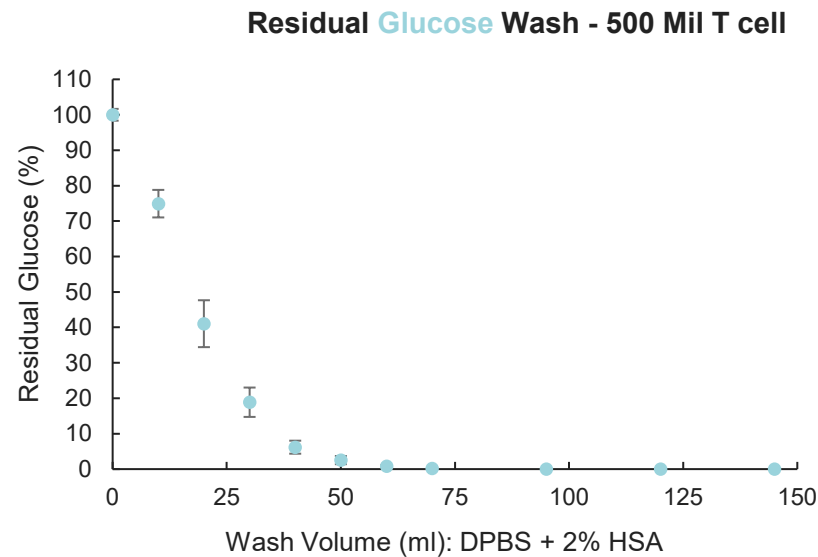
60mL Wash



70mL Wash

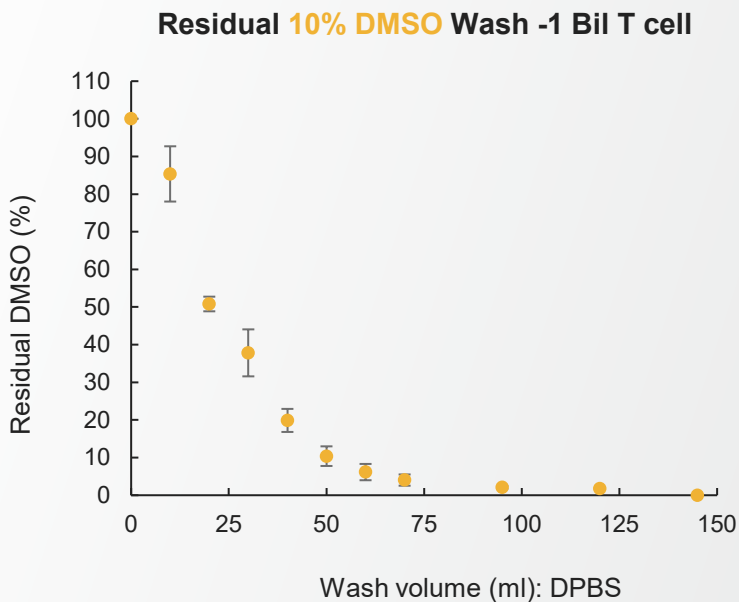


Glucose and DMSO Residuals Washout



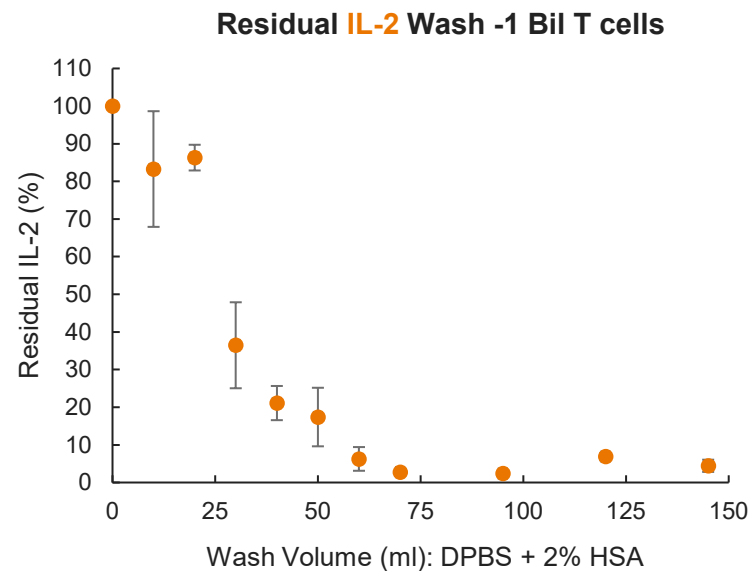
Wash Volume (mL)	Removed Glucose (%)
0	0%
10	25.08%
20	58.96%
30	81.12%
40	93.81%
50	97.45%
60	99.19%
70	99.79%
95	99.98%
120	*100.00%

* Level comparable to negative control

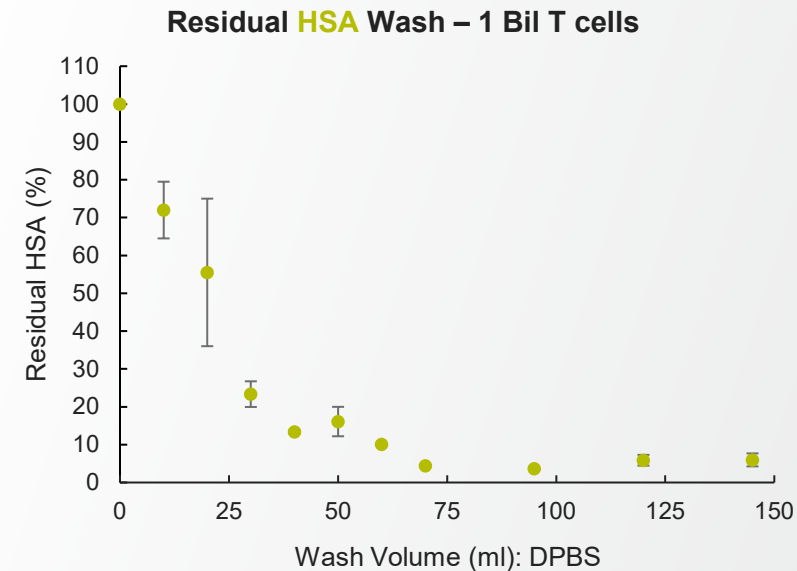


Wash Volume (mL)	Removed DMSO (%)
0	0%
10	14.64%
20	49.19%
30	62.18%
40	80.12%
50	89.62%
60	93.85%
70	95.97%
95	97.91%
120	98.21%
145	* 100%

IL-2 and HSA Residuals Washout

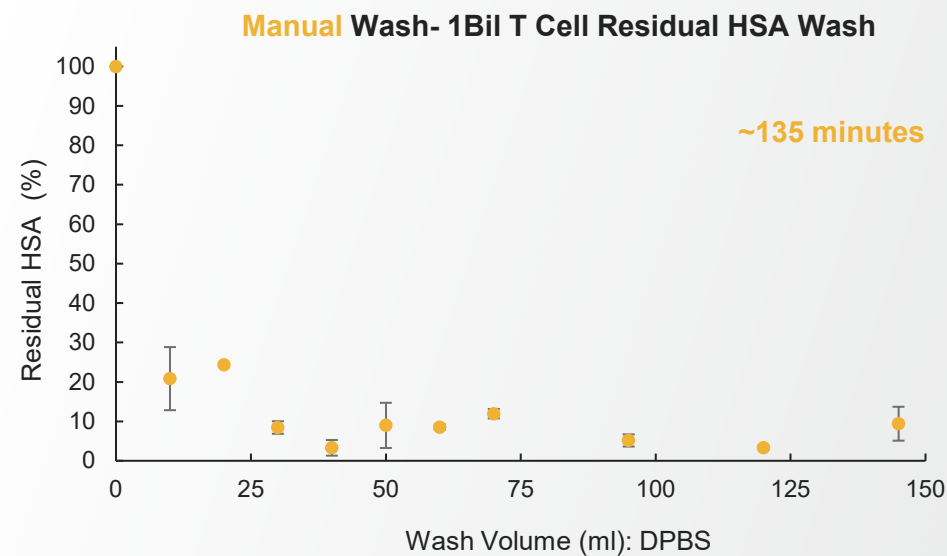
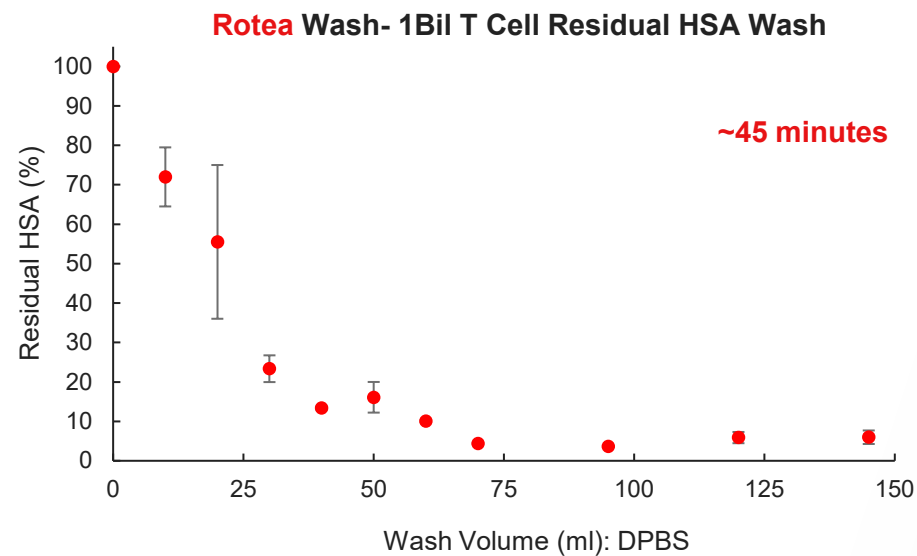


Wash Volume (mL)	Removed IL-2 (%)
0	0 %
10	16.72%
20	13.68%
30	63.54%
40	78.89%
50	82.61%
60	93.73%
70	97.25%
95	97.62%
120	93.11%
145	98.78%



Wash Volume (mL)	Removed HSA (%)
0	0%
10	28.01%
20	44.49%
30	76.65%
40	86.64%
50	83.89%
60	89.94%
70	95.62%
95	96.35%
120	94.11%
145	94.02%

Rotea vs manual HSA Residual Washout



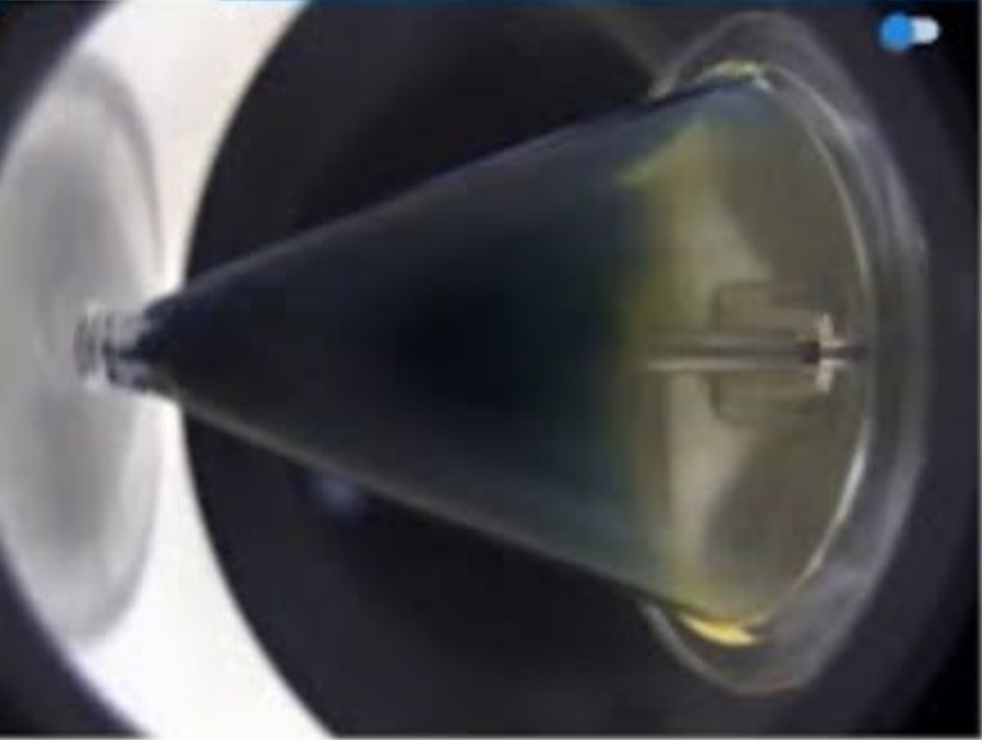
Wash Volume (mL)	Rotea Removed HSA (%)	Manual Removed HSA (%)
0	0.00%	0.00%
10	28.01%	79.16%
20	44.49%	75.67%
30	76.65%	91.56%
40	86.64%	96.71%
50	83.89%	91.03%
60	89.94%	91.47%
70	95.62%	88.05%
95	96.36%	94.85%
120	94.11%	96.78%
145	94.02%	90.59%

Full Bead Kit Demo Video

Elutriation of Yellow Beads

Live Video

Kit View



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11. Add 2.5 ml Flow

12. Ramp Speeds to Distribute Yellow Beads

▶ 13. Elutriate Yellow Beads

14. Ramp Speeds to Stabilize Bed

15. Recover Blue Beads

Instrument Status: Running

Total Time Remaining

Step Time Remaining 33

Pump On/Off 55

Centrifuge (g) 2200

